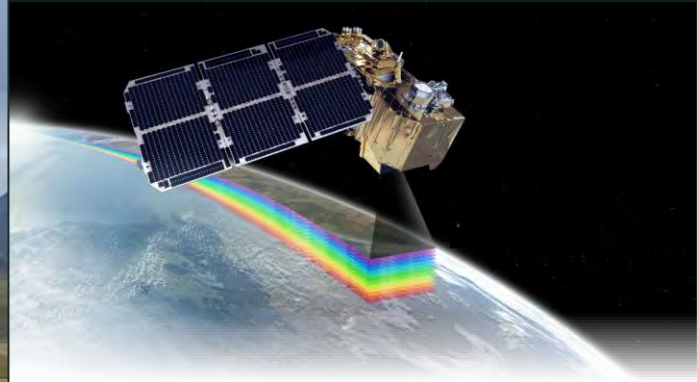






Algorithms for the Anthropocene: The Promise and Pitfalls of Machine Learning in a Changing Arctic

JOANMARIE DEL VECCHIO, ASSISTANT PROFESSOR, WILLIAM & MARY, WILLIAMSBURG, VA

With contributions from Kibiwott Koech, Aayla Kastning, Addison Greenfield, Noah Rupert, Samantha Gaylor, Gabriel Imholte



Geomorphologists translate between the languages of **landscapes** and **first physical principles** to describe and predict topographic form and change

Process (physics)-based geomorphology provides framework for predicting climate effects beyond historical observations



Southern California, Susan Cannon, USGS



White River, Hartford VT, USGS



Beaufort Sea, Alaska, USGS

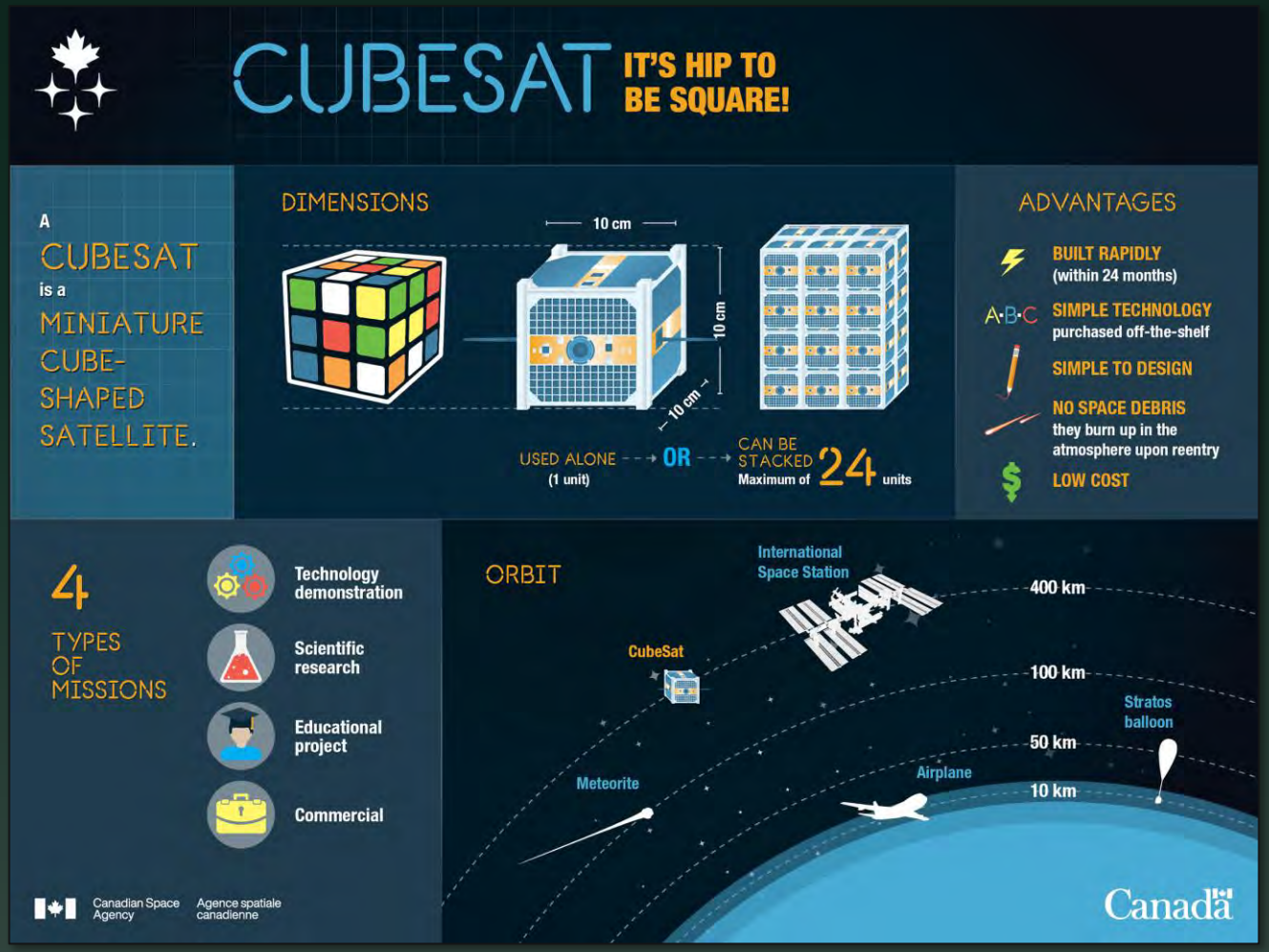


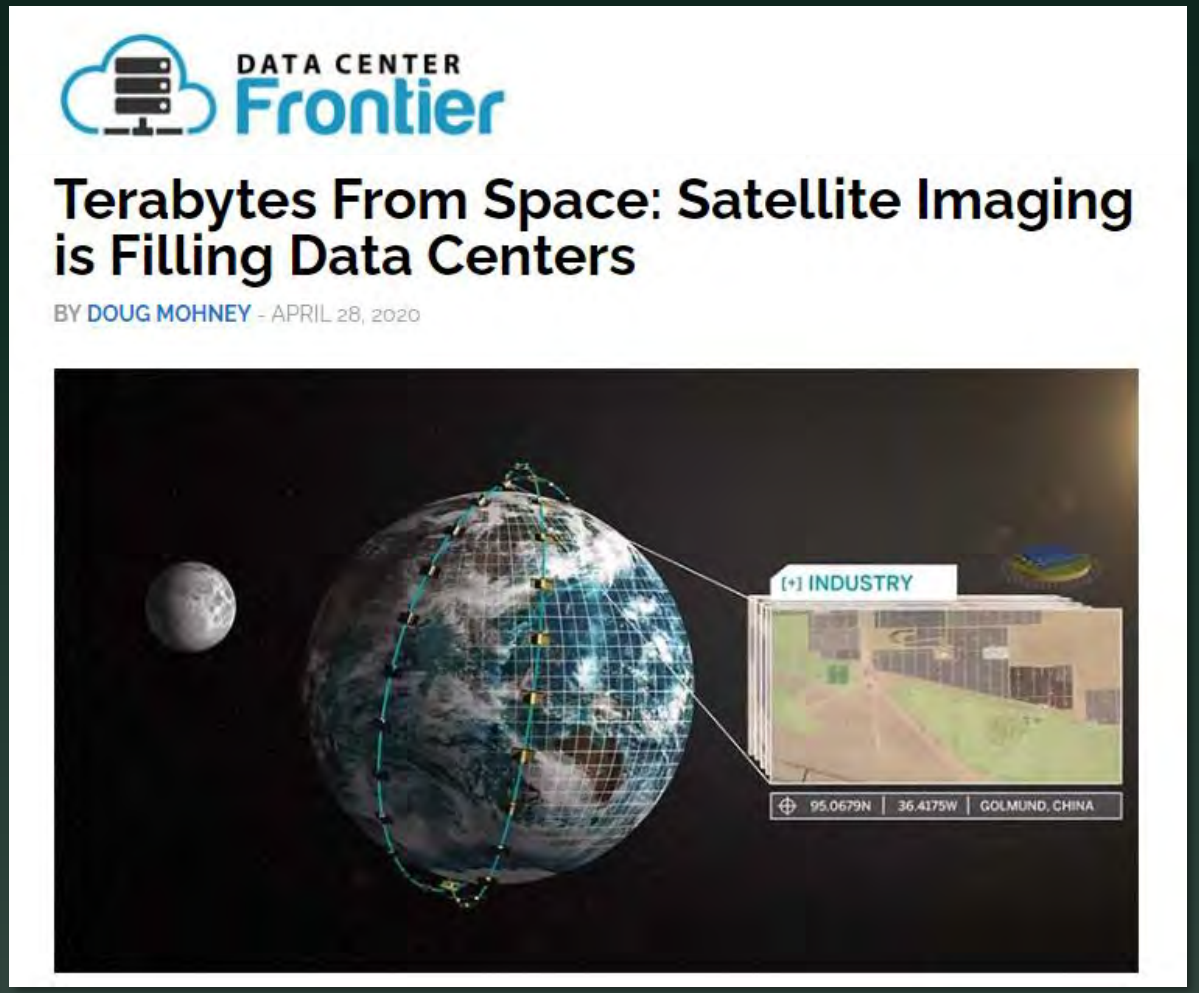
By Claudius Ptolemaeus

Better data
Better computers
Better algorithms



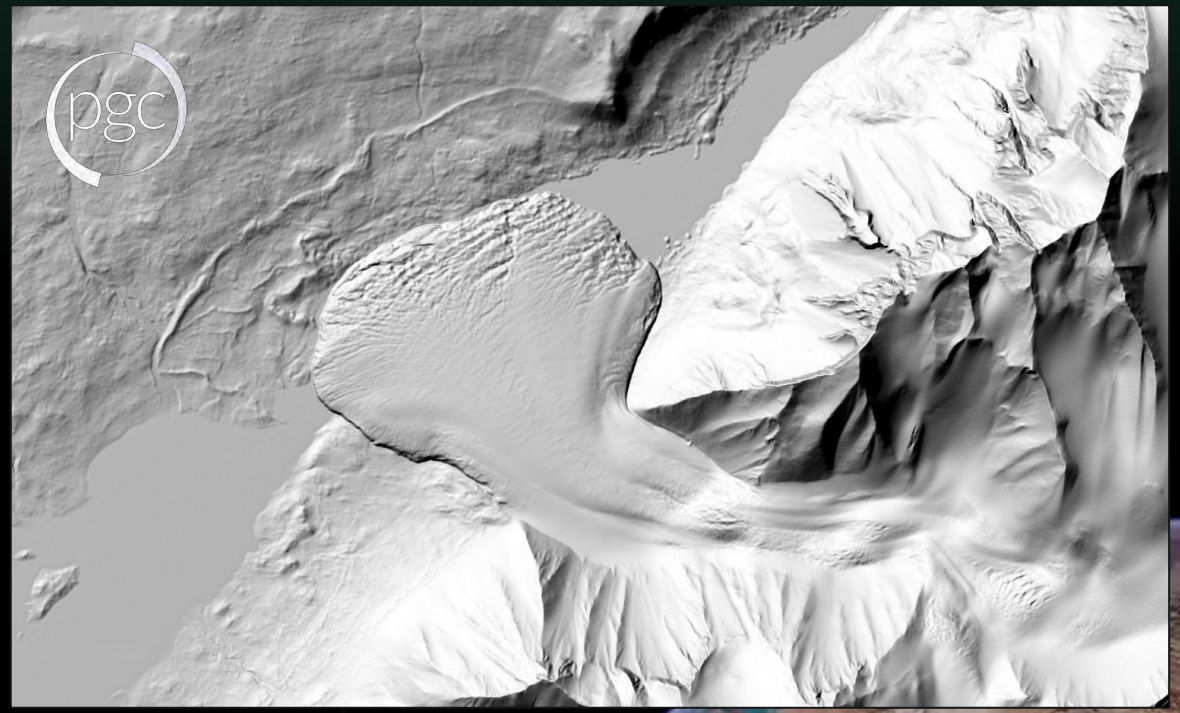
San Francisco Bay Area as seen by NASA Earth Resources Technology Satellite (ERTS) Ref E-1075-18178-3 02



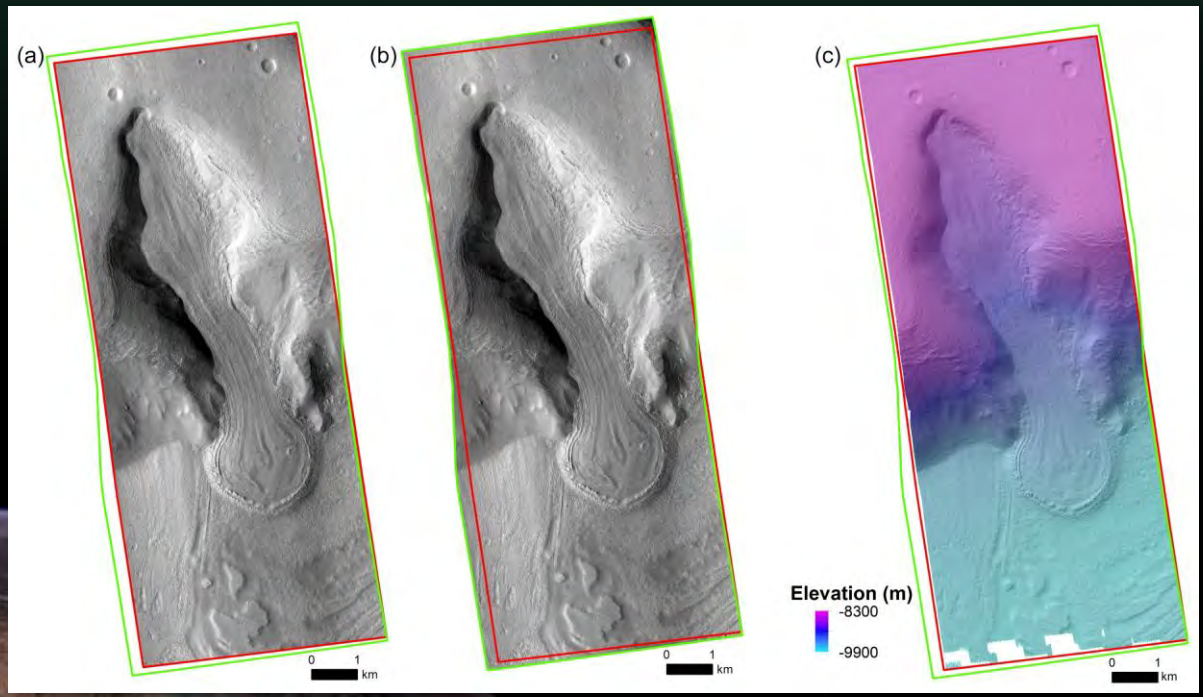


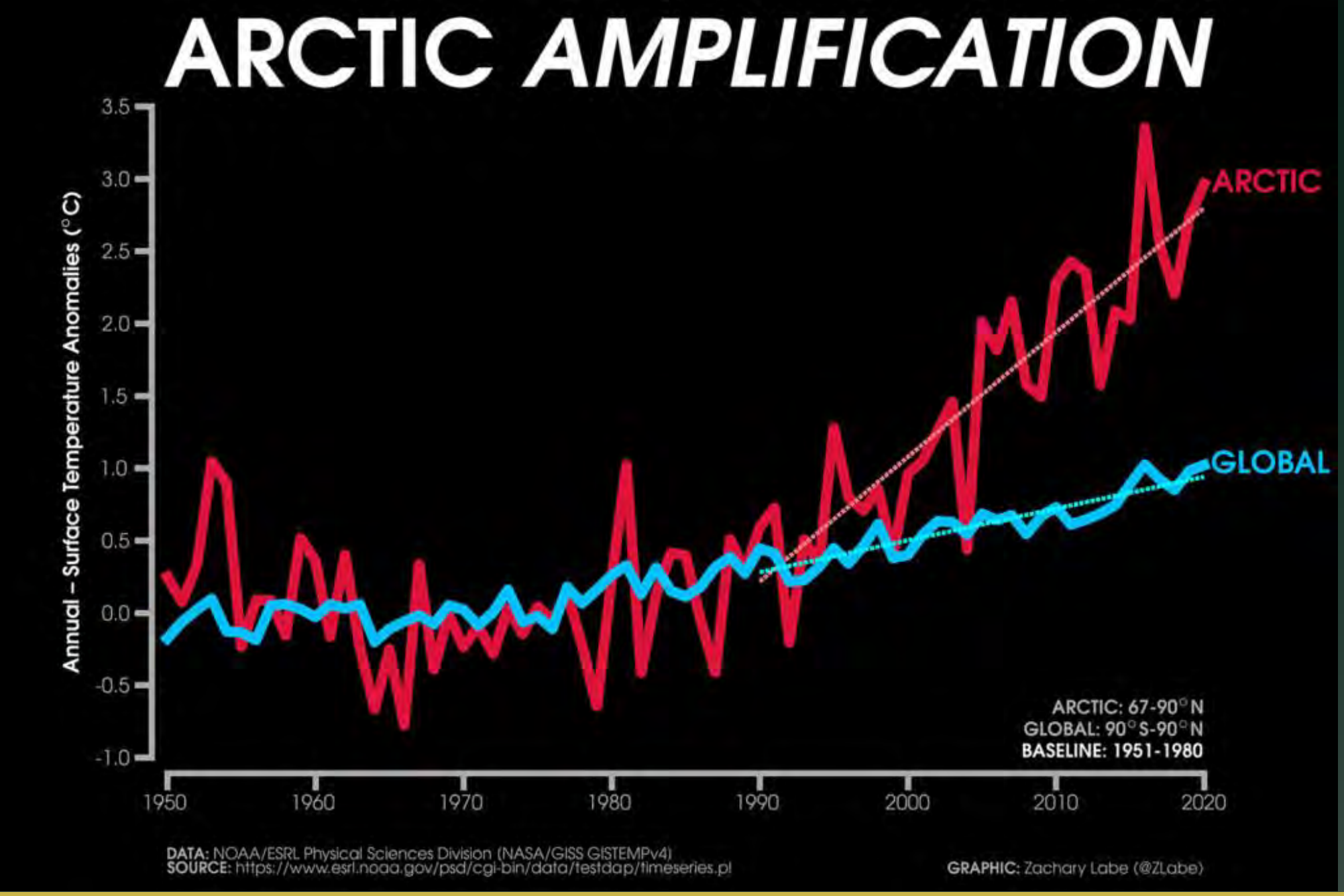
CubeSat at a glance. (Credit: Canadian Space Agency)

2 m digital elevation models of **poles** in **2018**

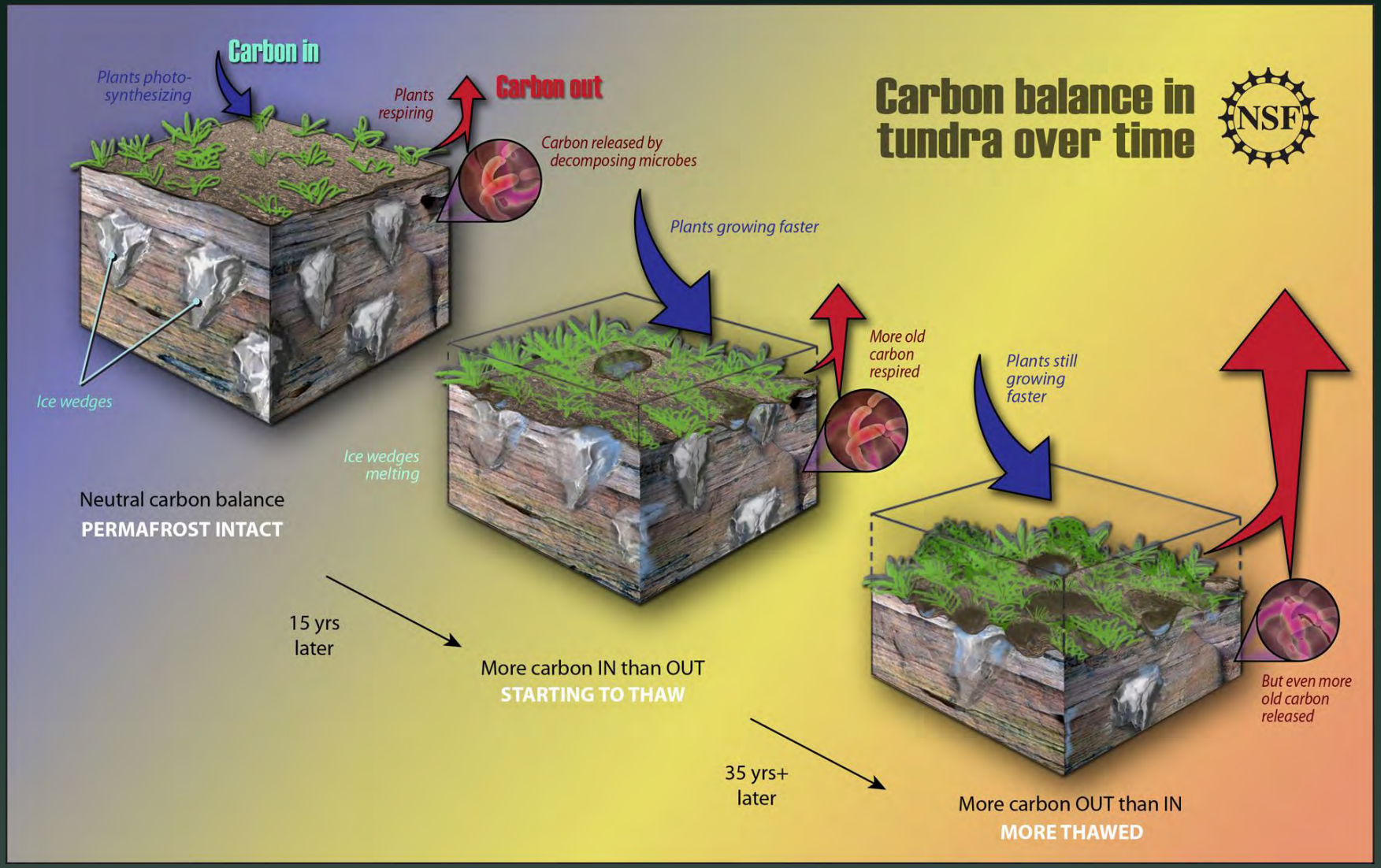


1 m digital elevation models of **Mars** in **2008**





Annual average surface air temperature anomalies for the Arctic (red line) and the global average (blue) from 1950 to 2020. Linear trend lines (dashed) are also shown over the 1990 to 2020 period. Credit: Data from GISTEMPv4, chart by Zack Labe.



Carbon Balance in the Tundra Over Time
Courtesy Zina Deretsky, National Science Foundation

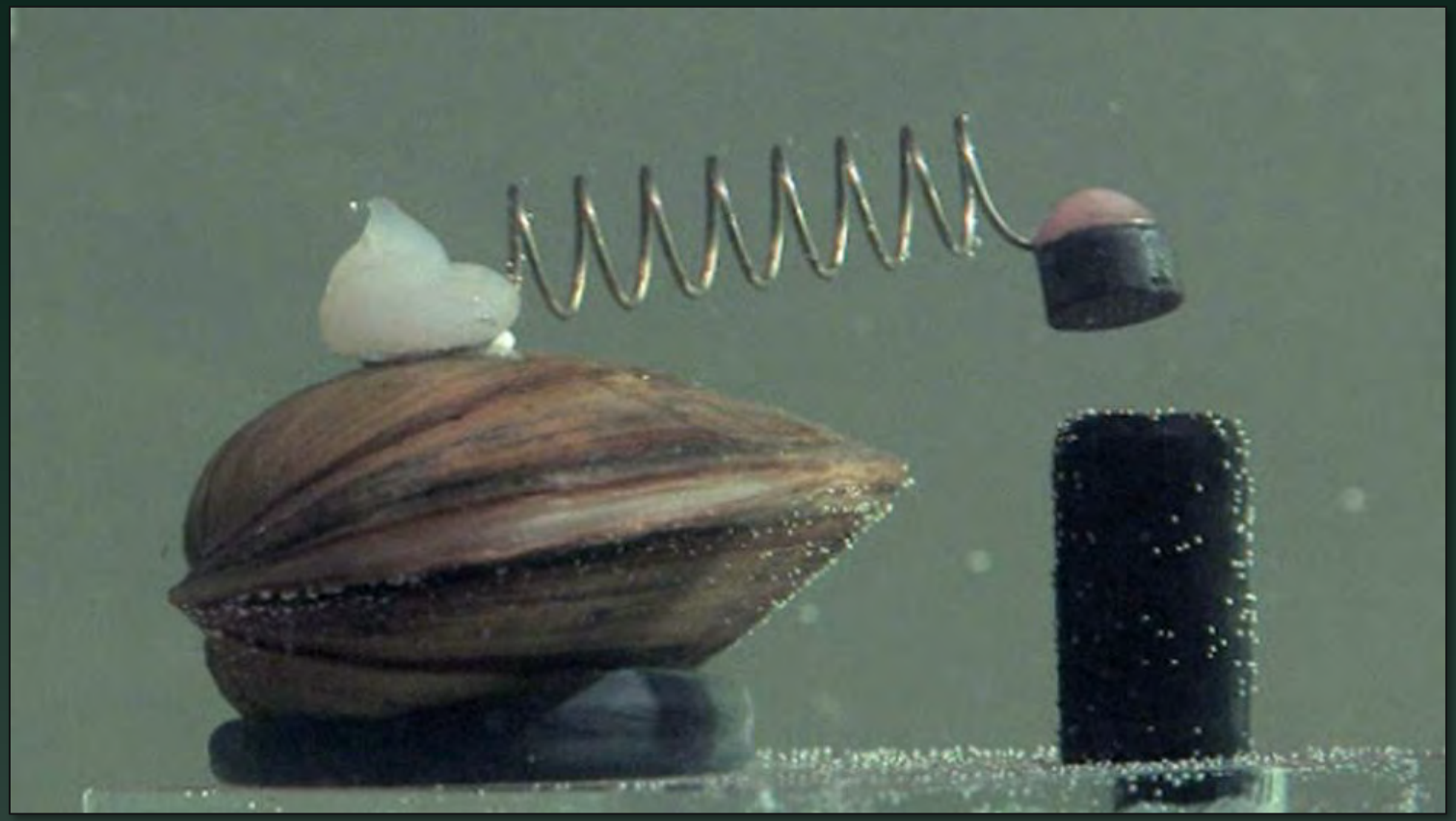


NPS/ Neal Herbert

In the Arctic, remote characterization is the **rule** rather than the **exception**.



Long before a miner feels the effect of gas in a mine a canary bird suffocates. That's why canaries are used as gas-alarms



Sentinel species as indicators of environmental hazards

BRITISH ECOLOGICAL SOCIETY Journal of Applied Ecology

RESEARCH ARTICLE | Free Access

The capacity of sentinel species to detect changes in environmental conditions and ecosystem structure

T. J. Clark-Wolf, Katie A. Holt, Erik Johansson, Anna C. Nisi, Kasim Rafiq, Leigh West, P. Dee Boersma, Elliott L. Hazen, Sue E. Moore, Briana Abrahms

First published: 07 May 2024 | https://doi.org/10.1111/1365-2664.14669 | VIEW METRICS

"…indicators of human impacts and ecosystem changes **that are otherwise difficult or impossible to measure**"

Science

Current Issue First release papers Archive About Submit manuscript

HOME › SCIENCE › VOL. 387, NO. 6735 › ELEPHANT SEALS AS ECOSYSTEM SENTINELS FOR THE NORTHEAST PACIFIC OCEAN TWILIGHT ZONE

RESEARCH ARTICLE MARINE CONSERVATION

Elephant seals as ecosystem sentinels for the north-east Pacific Ocean twilight zone

ROXANNE S. BELTRAN, ALLISON R. PAYNE, A. MARM KILPATRICK, CONNER M. HALE, MADISON REED, ELLIOTT L. HAZEN, STEVEN J. BOGRAD, JOFFREY JOUMA'A, PATRICK W. ROBINSON, [...], AND DANIEL P. COSTA +14 authors Authors Info & Affiliations

SCIENCE · 13 Feb 2025 · Vol 387, Issue 6735 · pp. 764-769 · DOI: 10.1126/science.adp2244

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NPS/ Neal Herbert

What can Arctic landscape form and change tell us about the **thermal state** of permafrost?

Landscapes are **surface sentinels** for subsurface dynamics!



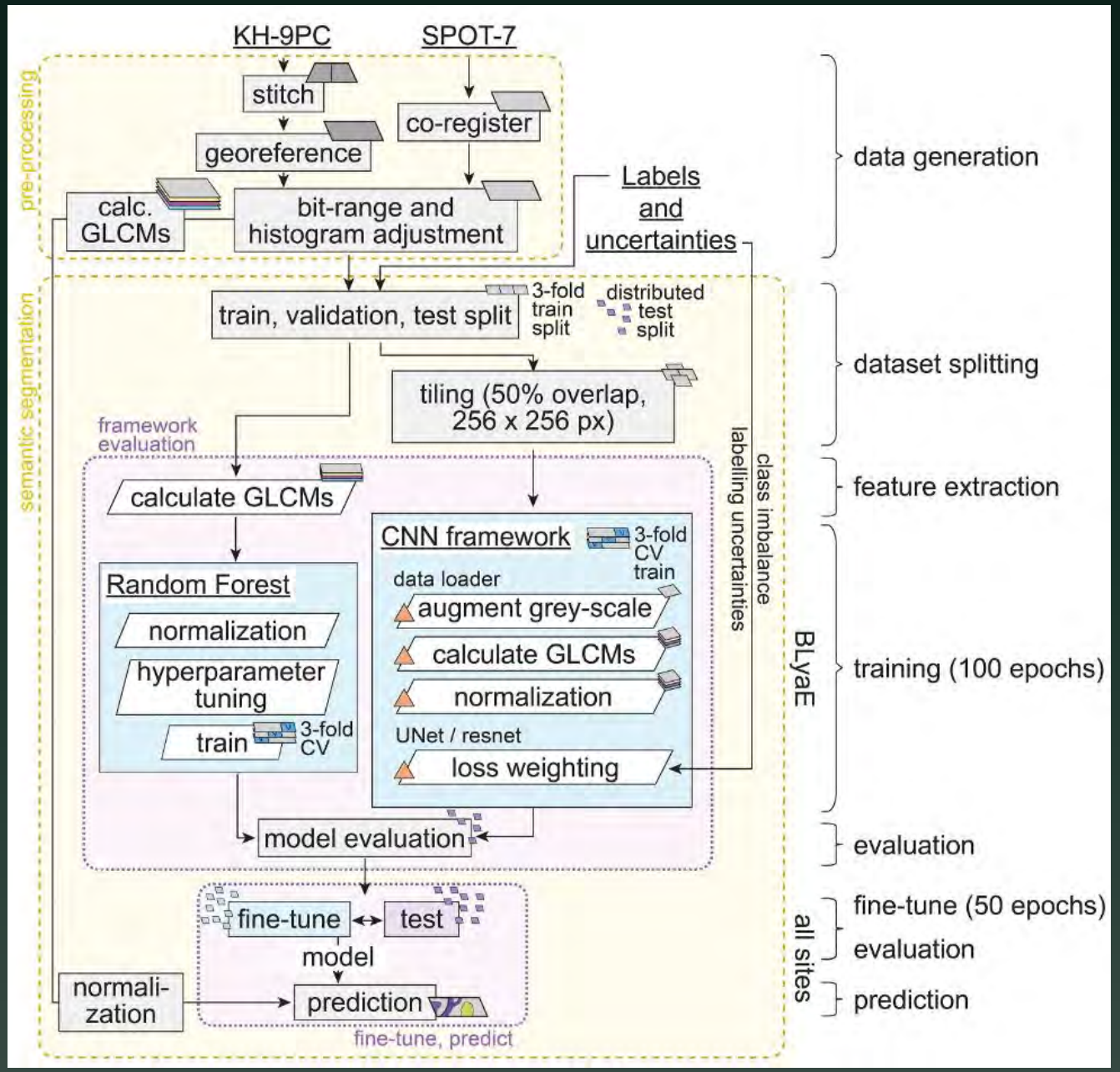
Permafrost thaw slump (Image courtesy: Melissa Ward Jones)

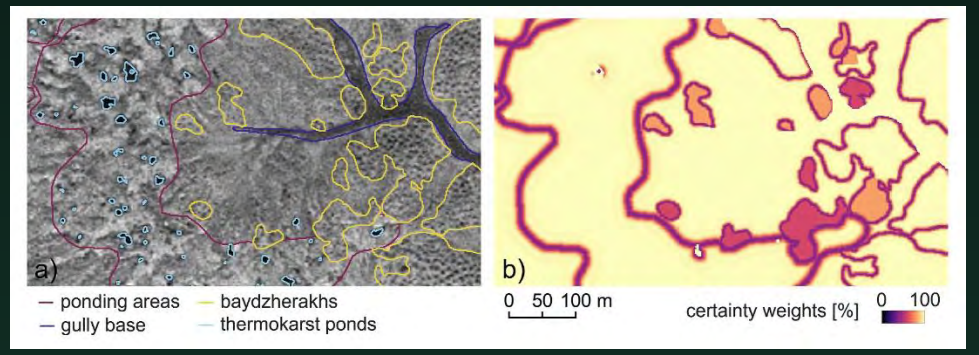


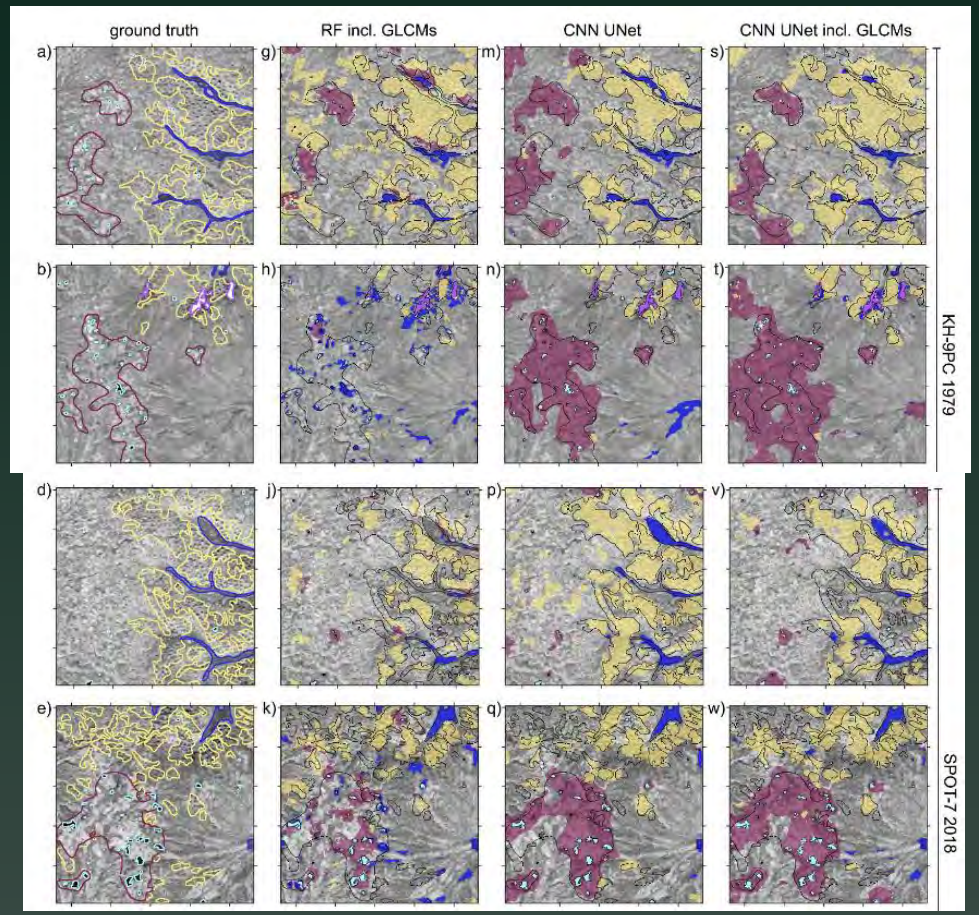
An aerial view of a large Arctic lake, nearly completely dry. Credit: David Swanson/National Park Service



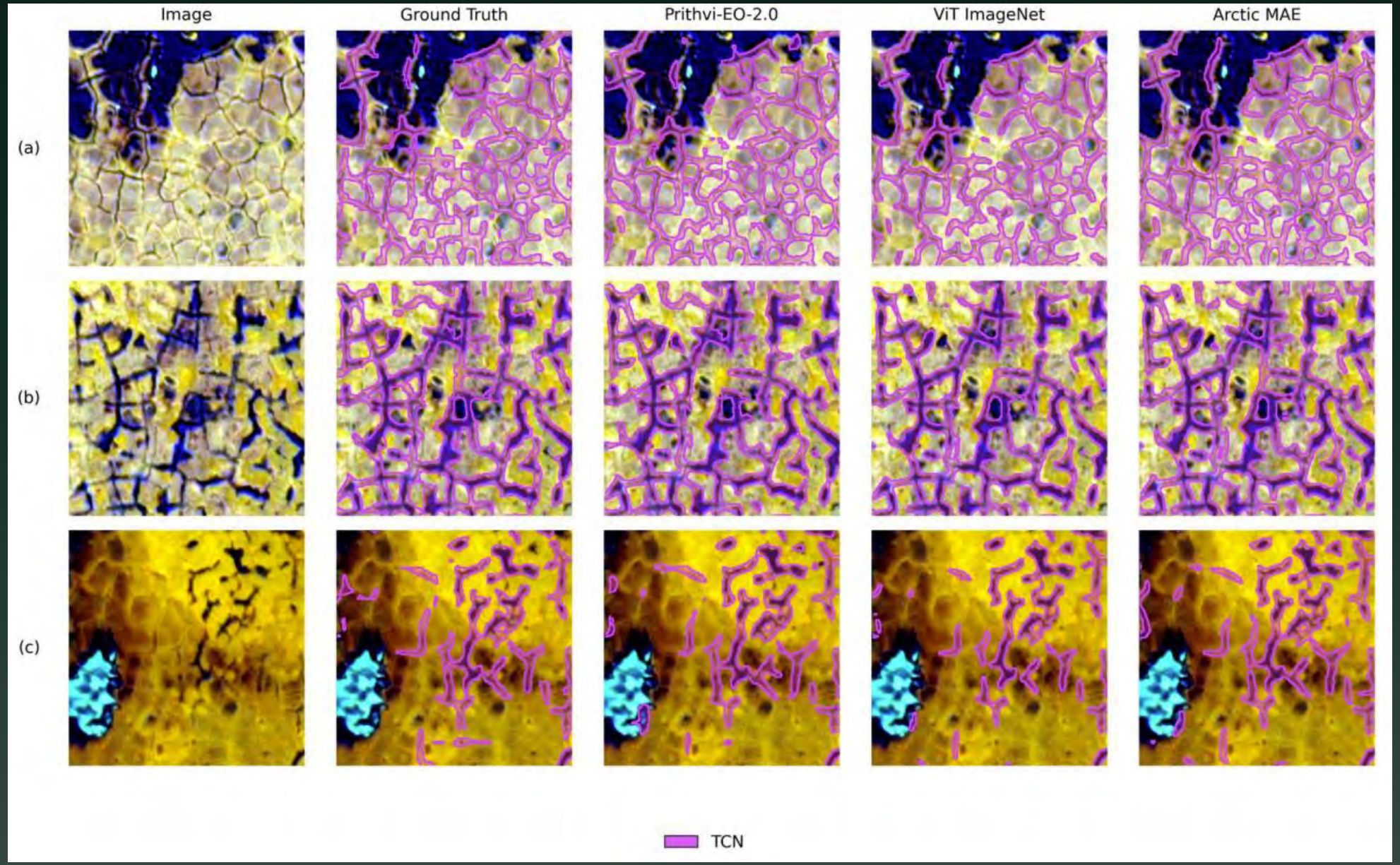
Josh Koch/U.S. Geological Survey



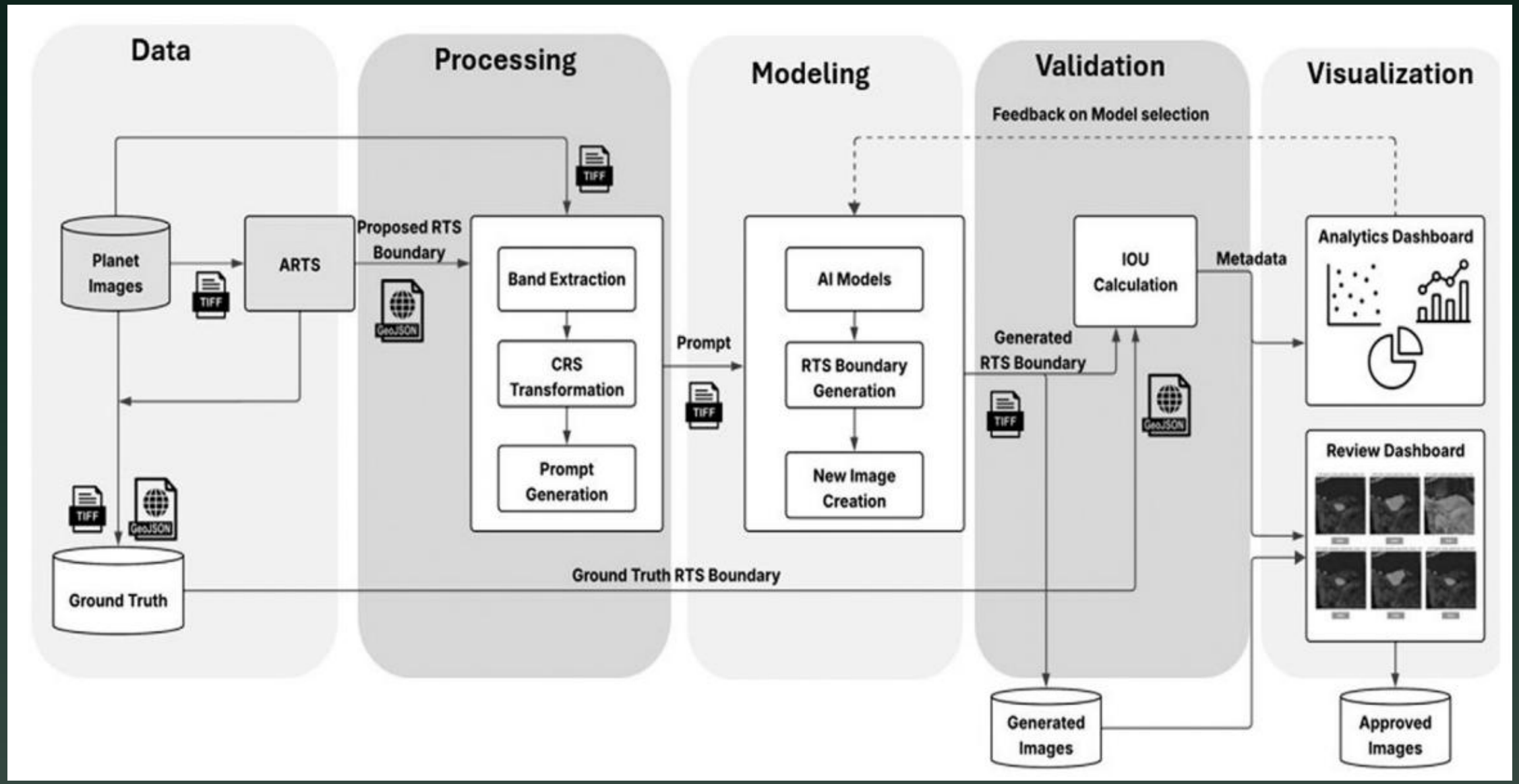




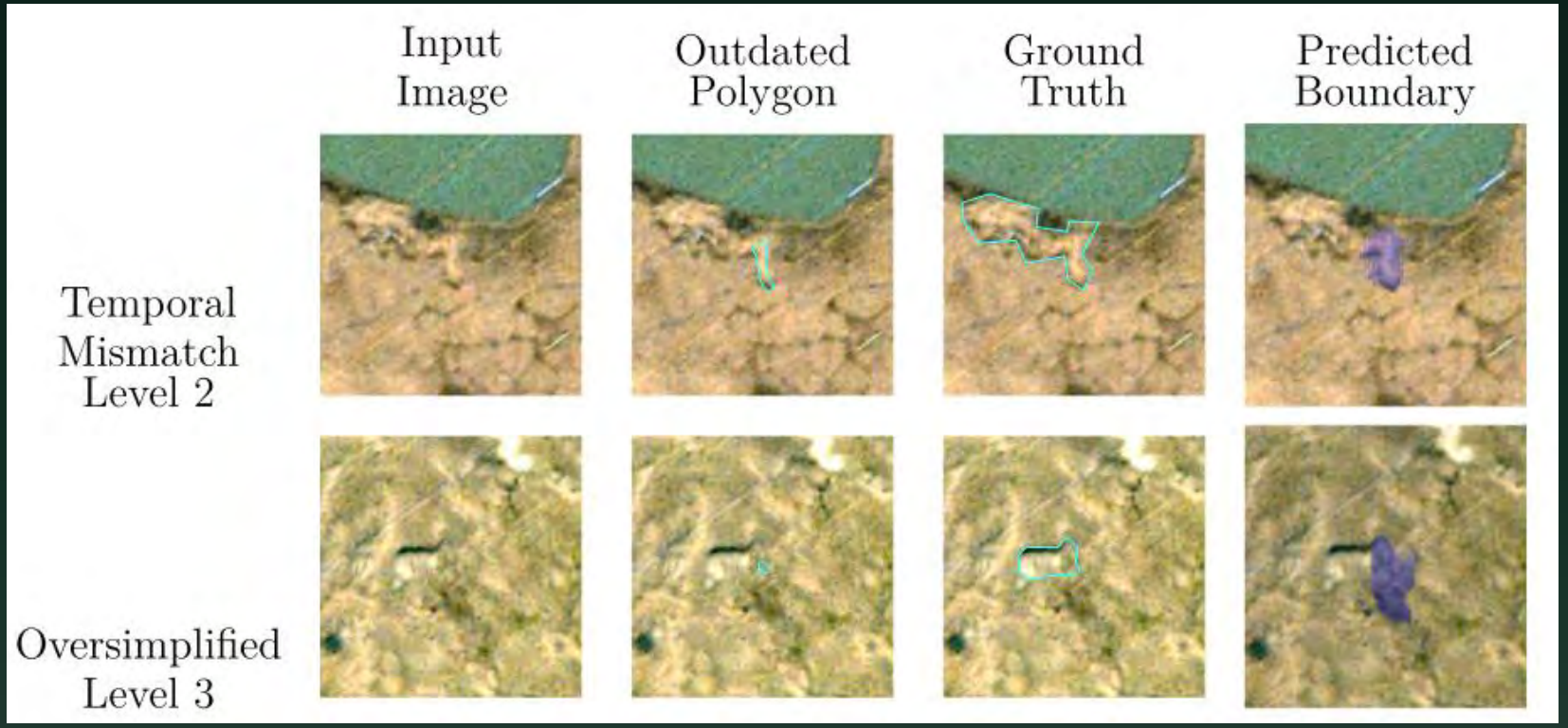
Inauen, C. M., Nitze, I., Langer, M.,Morgenstern, A., Hajnsek, I., & Grosse, G.(2025). Using texture-based image segmentation and machine learning with high-resolution satellite imagery to assess permafrost degradation landforms in the Russian High Arctic. Journal of Geophysical Research: Machine Learning and Computation, 2, e2024JH000550.



Perera, Amal S., et al. "Clustering Guided Domain-Specific Pretrained Foundation Model for Very High-Resolution Arctic Remote Sensing." *arXiv preprint arXiv:2605.30467* (2026).



Yang, Yili, et al. "Keeping pace with a changing planet: An interactive segmentation framework for refining delineations of dynamic Earth features with the Segment Anything Model." *International Journal of Applied Earth Observation and Geoinformation* 147 (2026): 105187.



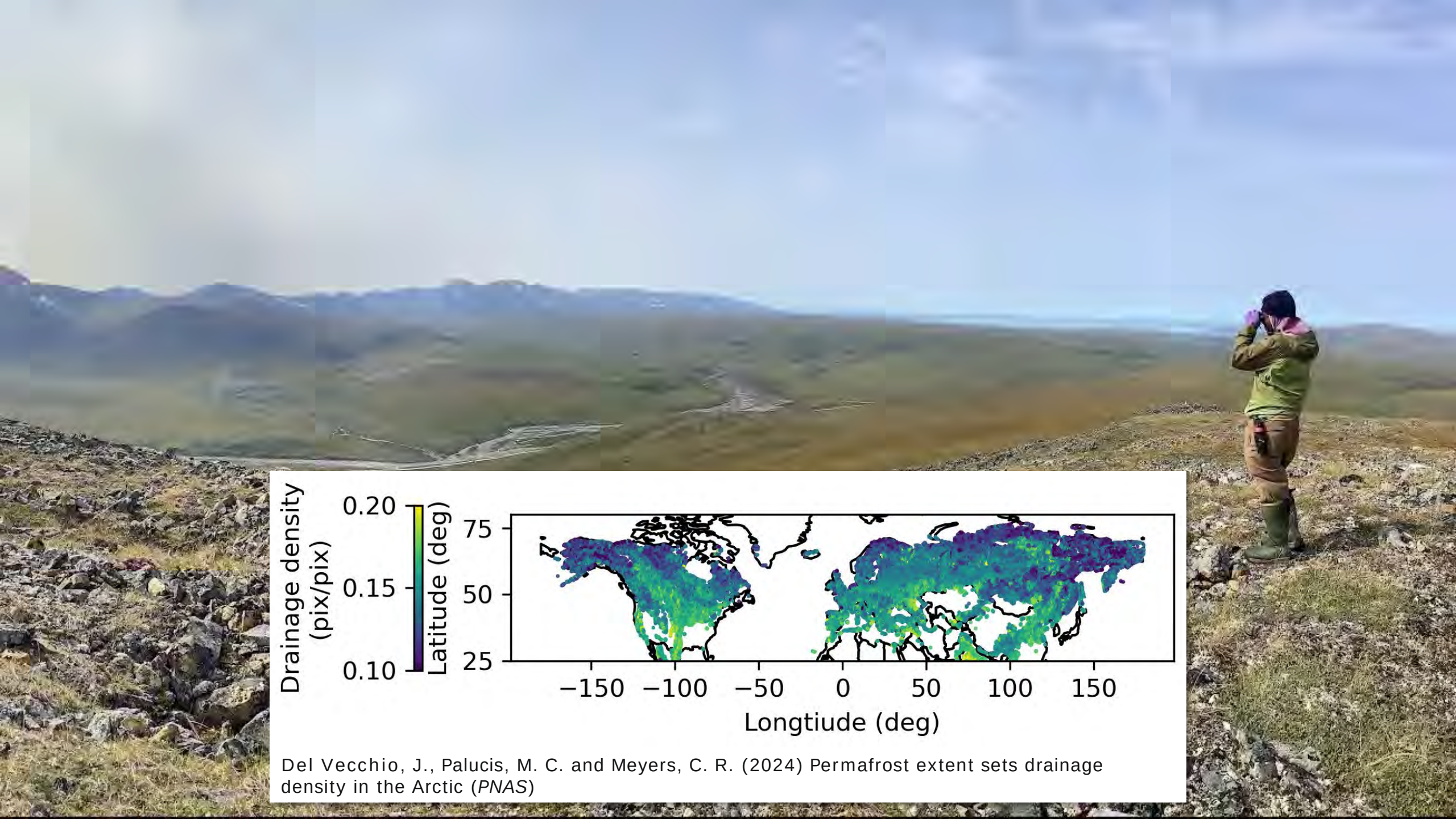
- Fixability Level (1–3):
  Indicates how difficult it is to correct the RTS boundary:

Level	Description
Level 1 (Easy)	The RTS polygon is mostly accurate and only requires minor adjustments (e.g., slight shift or a few added vertices).
Level 2 (Moderate)	The RTS is somewhat misaligned or simplified; requires moderate edits like shifting and adding several vertices.
Level 3 (Hard)	The RTS polygon is significantly incorrect. It's easier to redraw the entire polygon than to fix the existing one.

- Quality Issue Type (A–C):
  Describes the type of issue present in the original RTS polygon:

Code	Issue Type	Description
A	Spatial Mismatch	The RTS polygon is correctly shaped but **offset** from the actual feature in the imagery (e.g., shifted location).
B	Simplistic Mismatch	The polygon is correctly located but **overly simplified** (e.g., lacks sufficient detail or vertices).
C	Time Mismatch	The polygon was drawn using **older imagery**, so it's smaller or outdated compared to the current RTS footprint.

Yang, Yili, et al. "Keeping pace with a changing planet: An interactive segmentation framework for refining delineations of dynamic Earth features with the Segment Anything Model." *International Journal of Applied Earth Observation and Geoinformation* 147 (2026): 105187.



Del Vecchio, J., Palucis, M. C. and Meyers, C. R. (2024) Permafrost extent sets drainage density in the Arctic (*PNAS*)

Water tracks

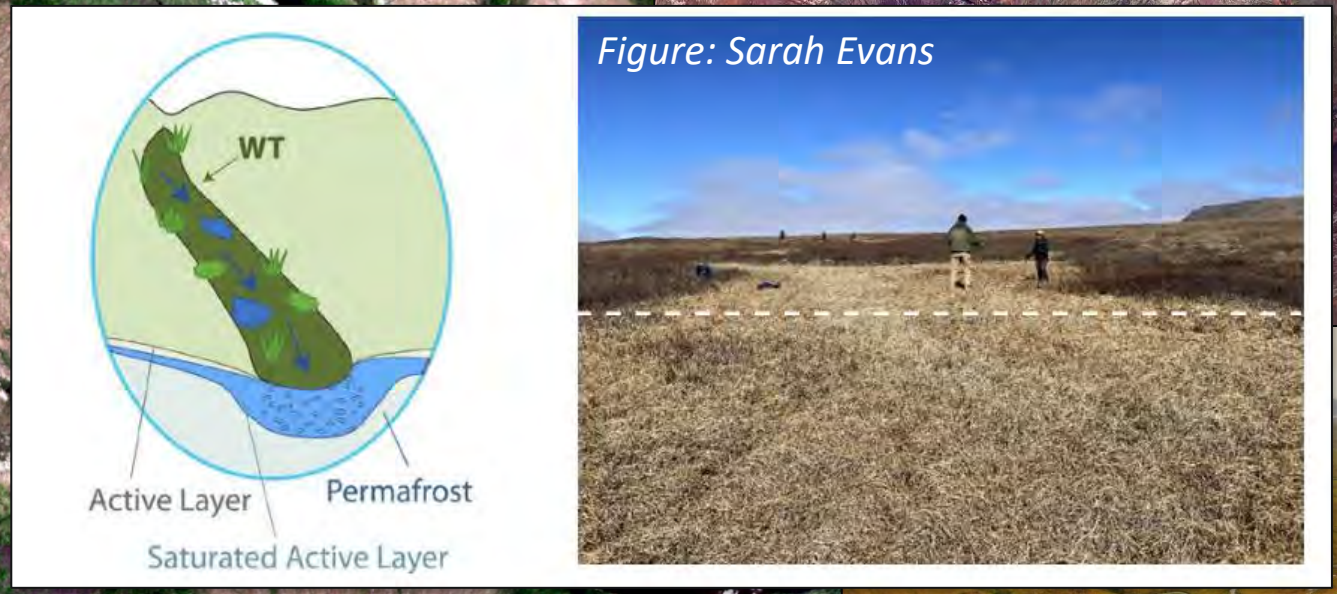


Photo credit: Qifan Yang.

Water tracks

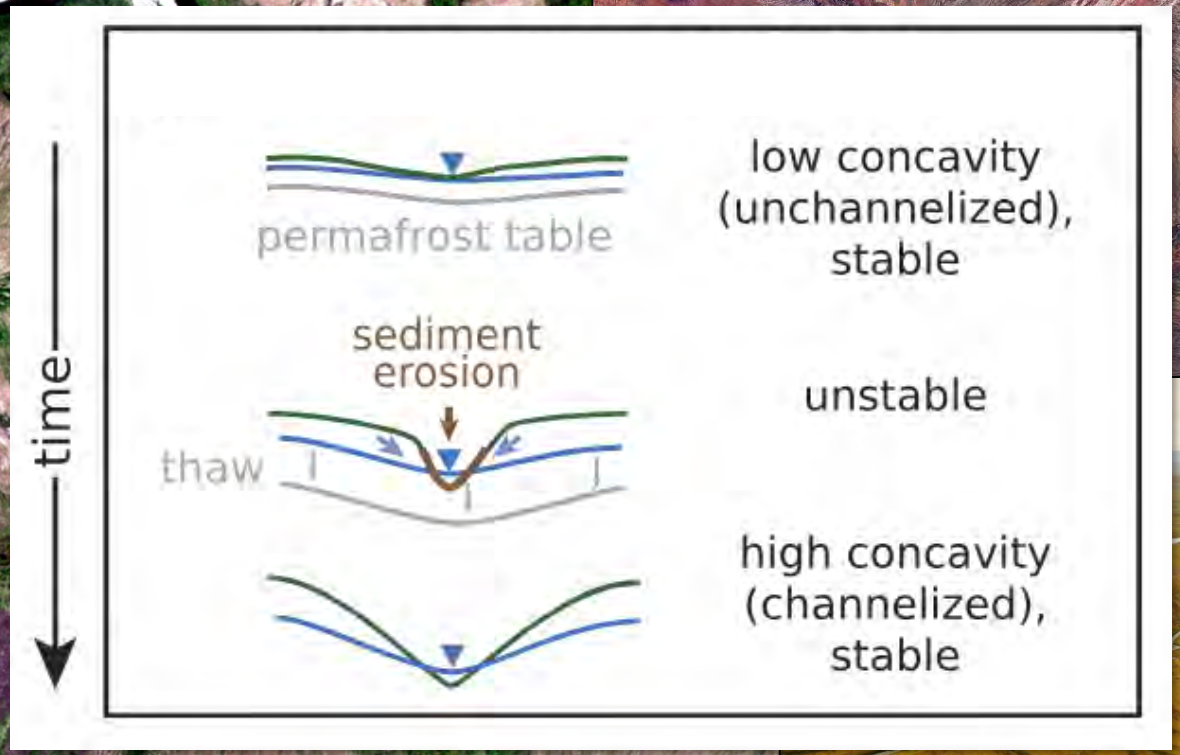
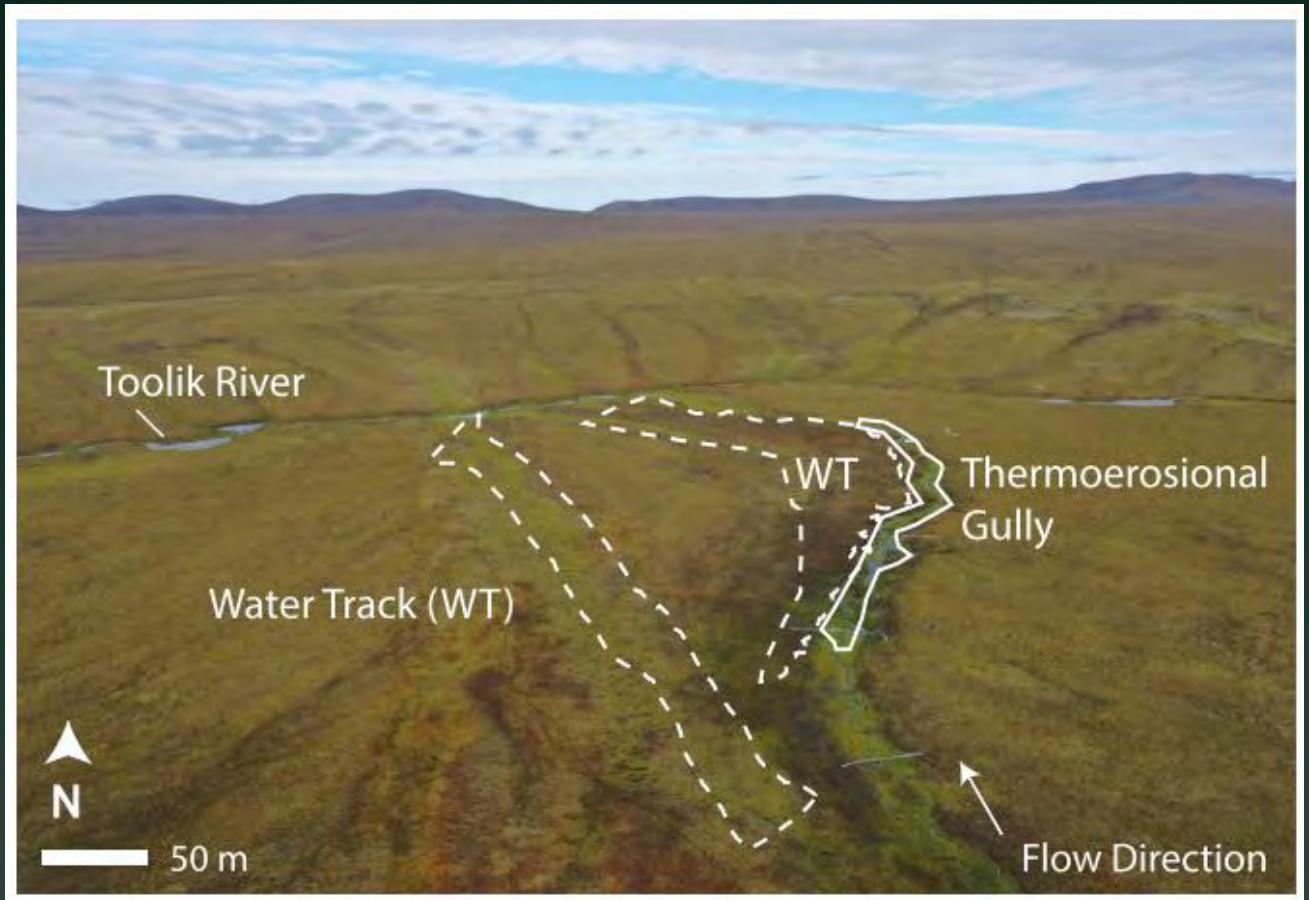
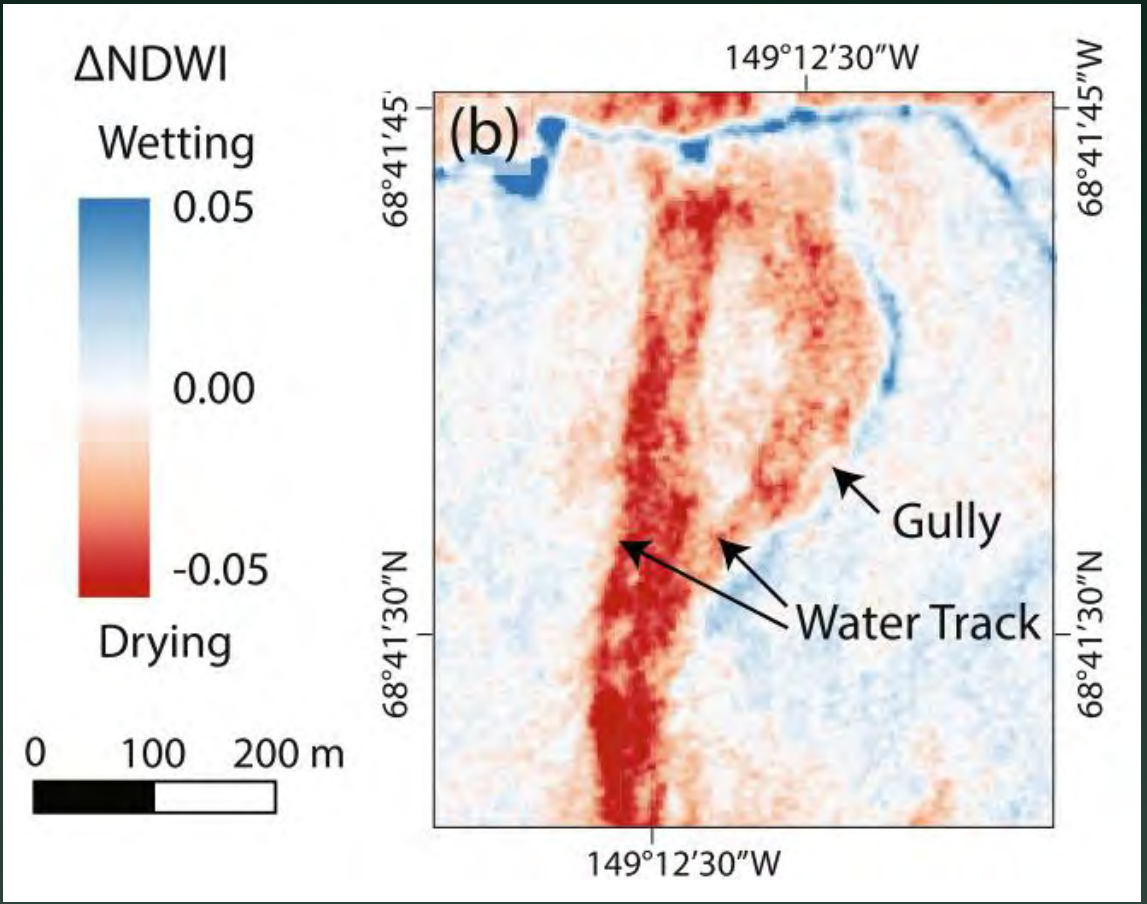


Photo credit: Qifan Yang.





Water tracks as excellent surface sentinels of subsurface dynamics!

Evans, S. G., C. C. Chew, and S. E. Godsey. 2025. " Near-Surface Drying of a Continuous Permafrost Hillslope With Water Tracks Following Ground Collapse." *Hydrological Processes* 39, no. 9: e70269. https://doi.org/10.1002/hyp.70269.

code.wm.edu/pycogss/PyCoGSS

Arctic landscapes are changing...and

PYCOGSS

(The **Py**thon **Co**mputational **G**eomorphology **S**oftware **S**ystem)

is here to help you and your students study them!

Where?
Define an area of interest (AOI) manually or by existing geometries

How?
Search spatio-temporal access catalogs (STACs) for overlapping multispectral, topographic, or other gridded data

When?
Perform simple to complex pixelwise time series analysis

Why?
Connect climate datasets, and/or perform geomorphic analyses of your choice

Now share!
Produce scalable, reproducable, interoperable observations

Created on
May 13, 2026



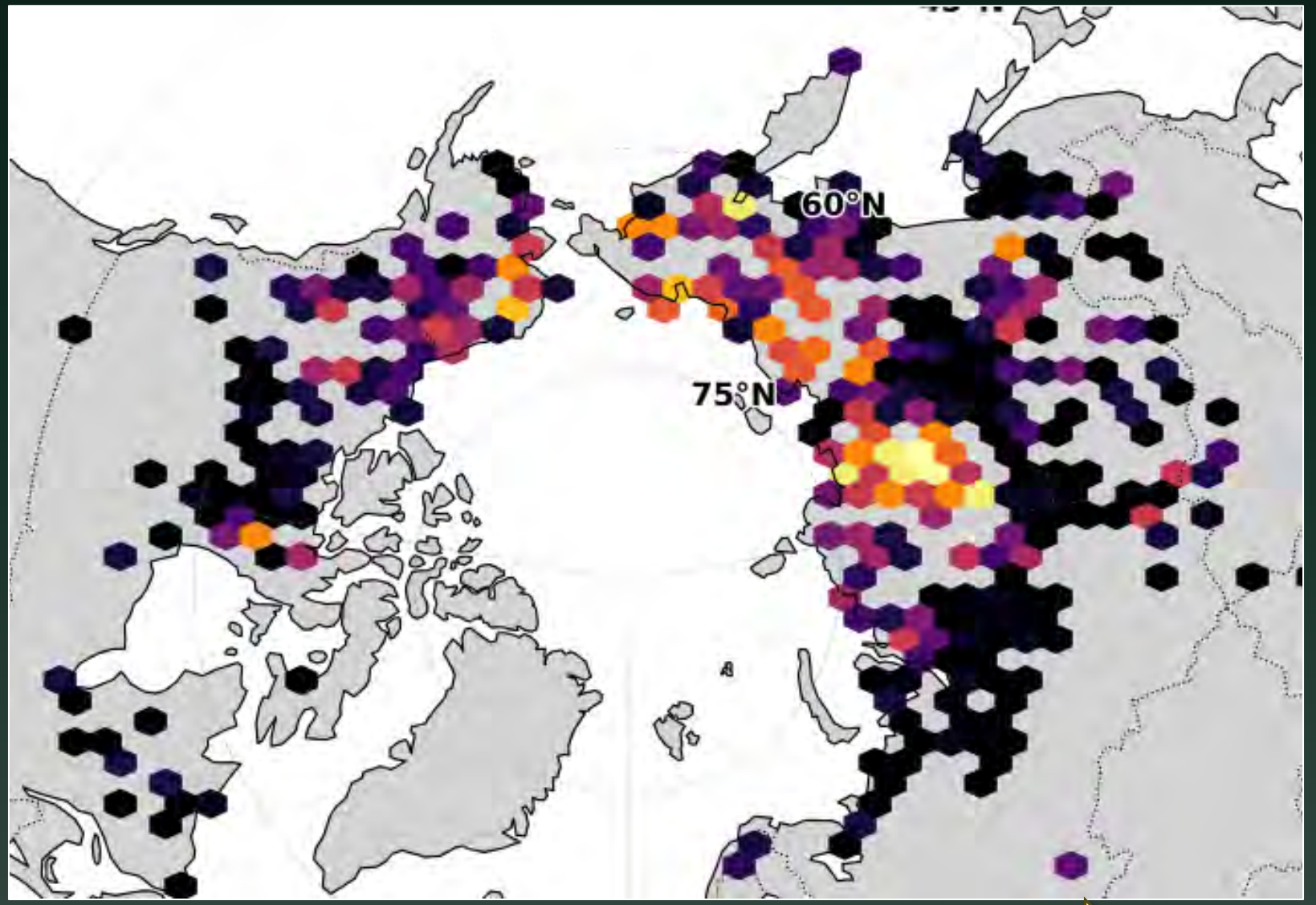
Kibiwott Koech
(applying for PhDs!)

Public dataset	Description	Sources	Available Bands/Variables	Resolution/Frequency
Sentinel-2 Level-2A	The global Sentinel-2 archive, from 2016 to the present, processed to L2A (bottom-of-atmosphere) using Sen2Cor algorithm and converted to cloud-optimized GeoTIFF format.	AWS Element84 OSDF Planetary Computer	`red`, `blue`, `green`, `nir`, `nir08`, `nir09`, `swir16`, `swir22`, `scl`, `cloud`, `snow` (Sentinel-2 bands), `ndvi`, `ndgi`, `ndsi`, `ndwi` (computed spectral indices, see Montero et al. [1]	10 meters (spatial); 2-8 days (temporal)
ERA5 Reanalysis	Global climate dataset for the period 1940-present, built through data assimilation (analysis and reanalysis), developed by ECWMF. More details here	Open-Meteo, NCAR	`temp_max`, `temp_min`, `rain_total`, `snowdepth`, `precip_total`	31km, daily
ArcticDEM	Digital surface model (DSM) of the Arctic derived from optical stereo imagery	UMN Polar Geospatial Center	`elevation`, `slope`, `curvature`, `aspect`	2 meters







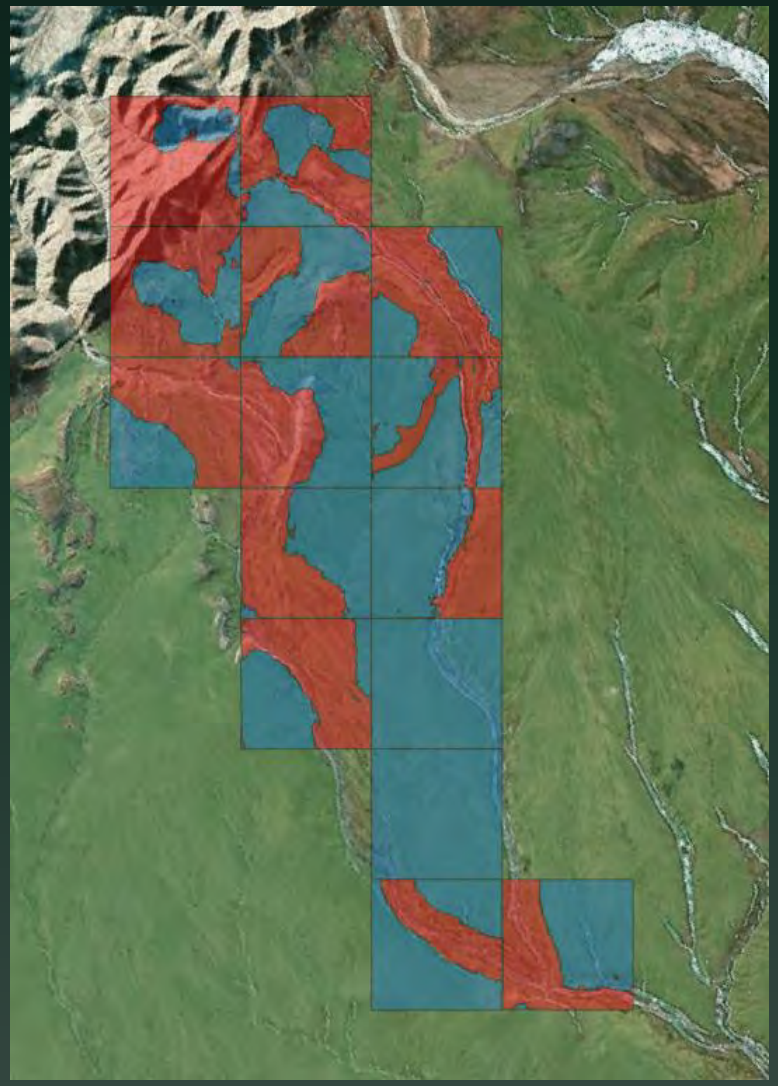


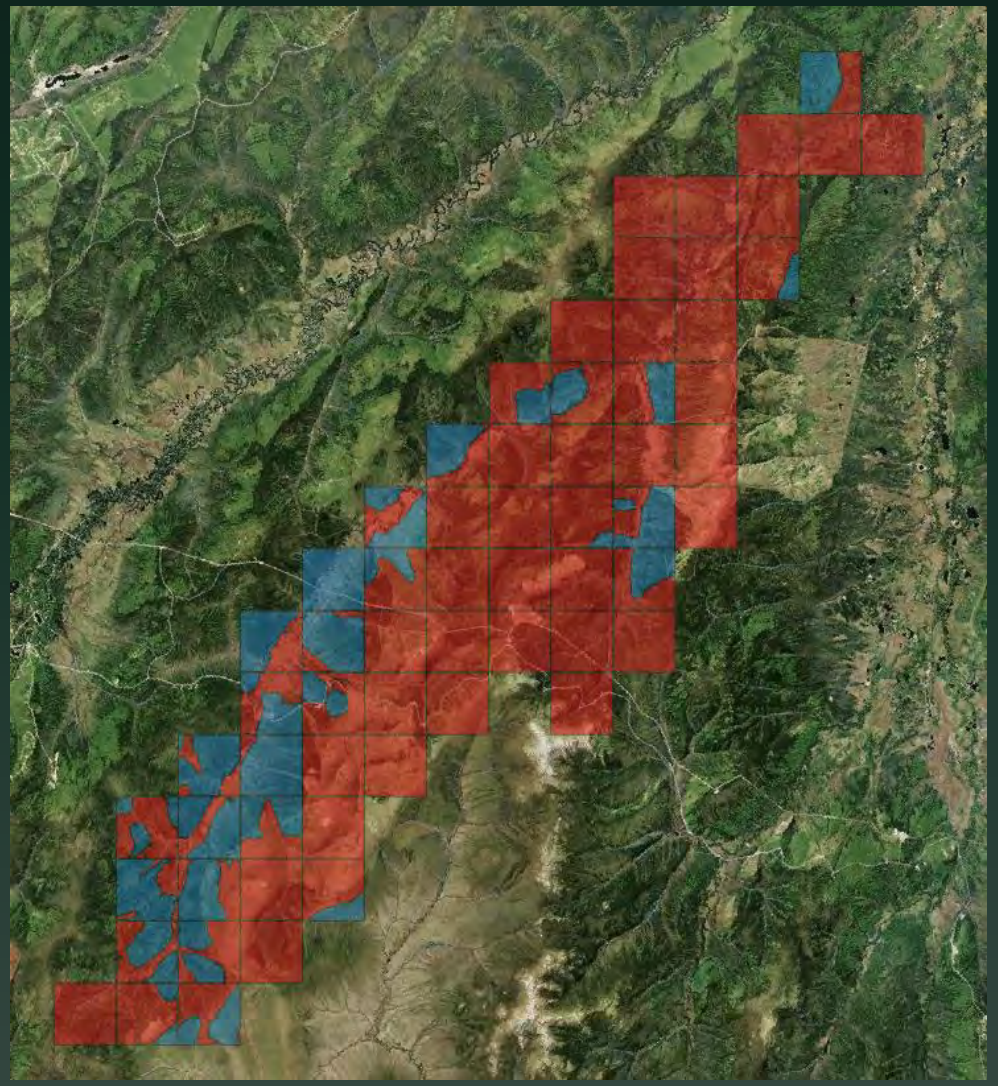
More water track-like features

850 watersheds

~32,000 km^2

0.17% of land underlain by permafrost

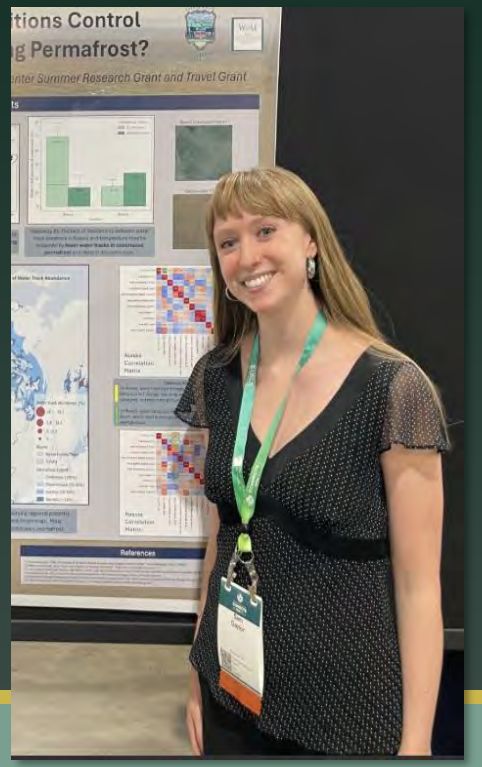




438 “tiles” of 256x256 10 m pixels

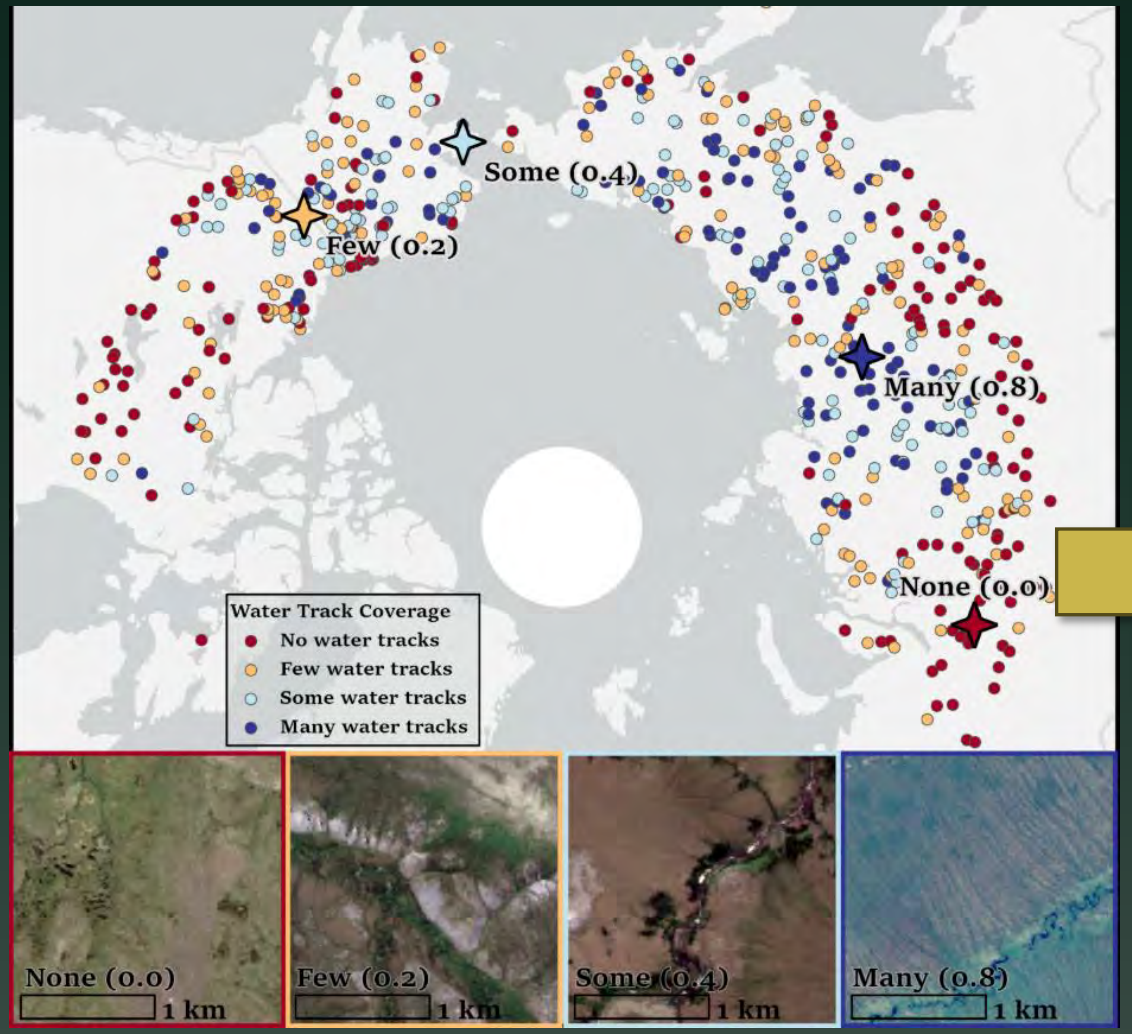
39 watersheds

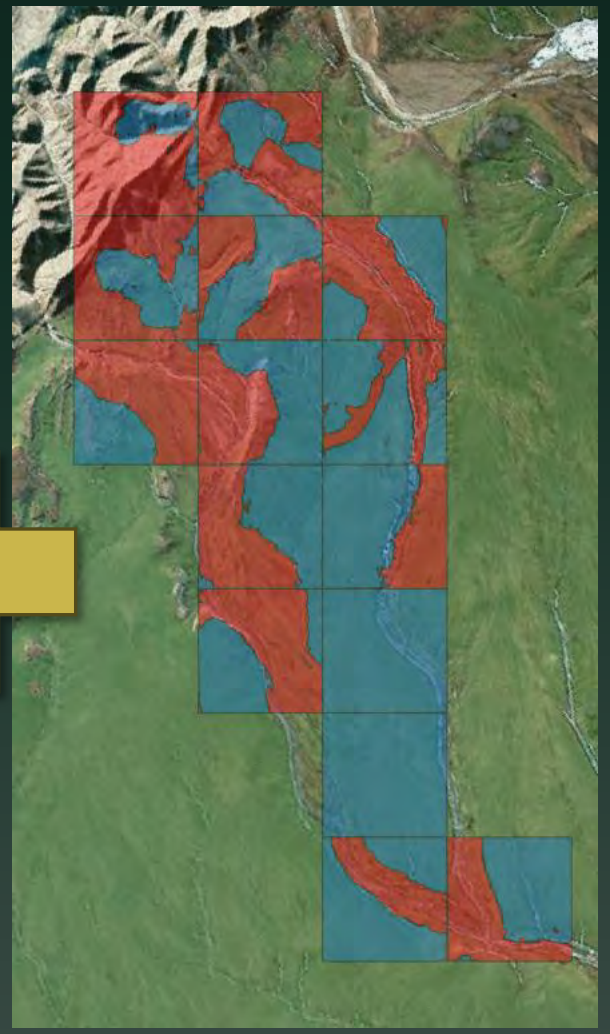
~2900 km^2



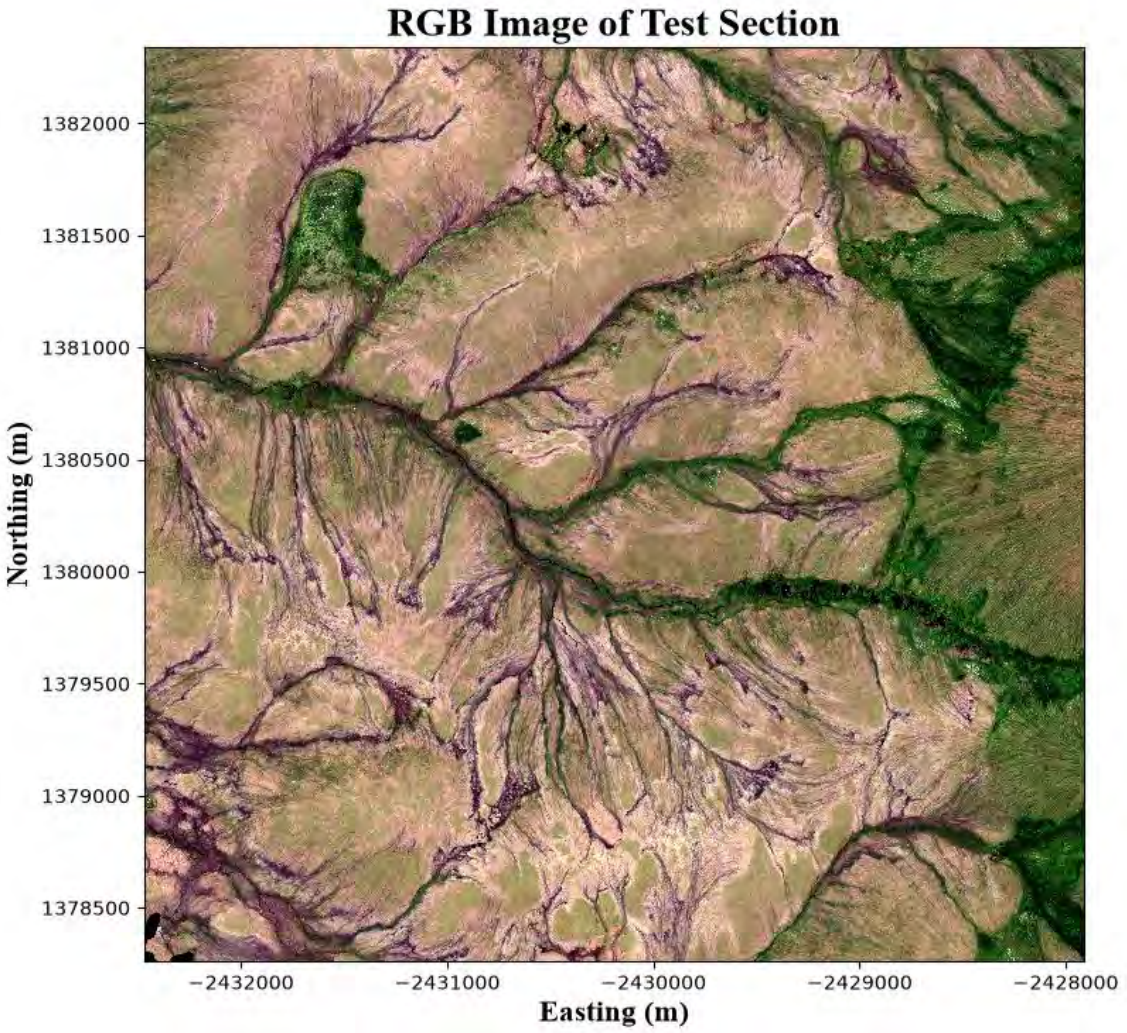


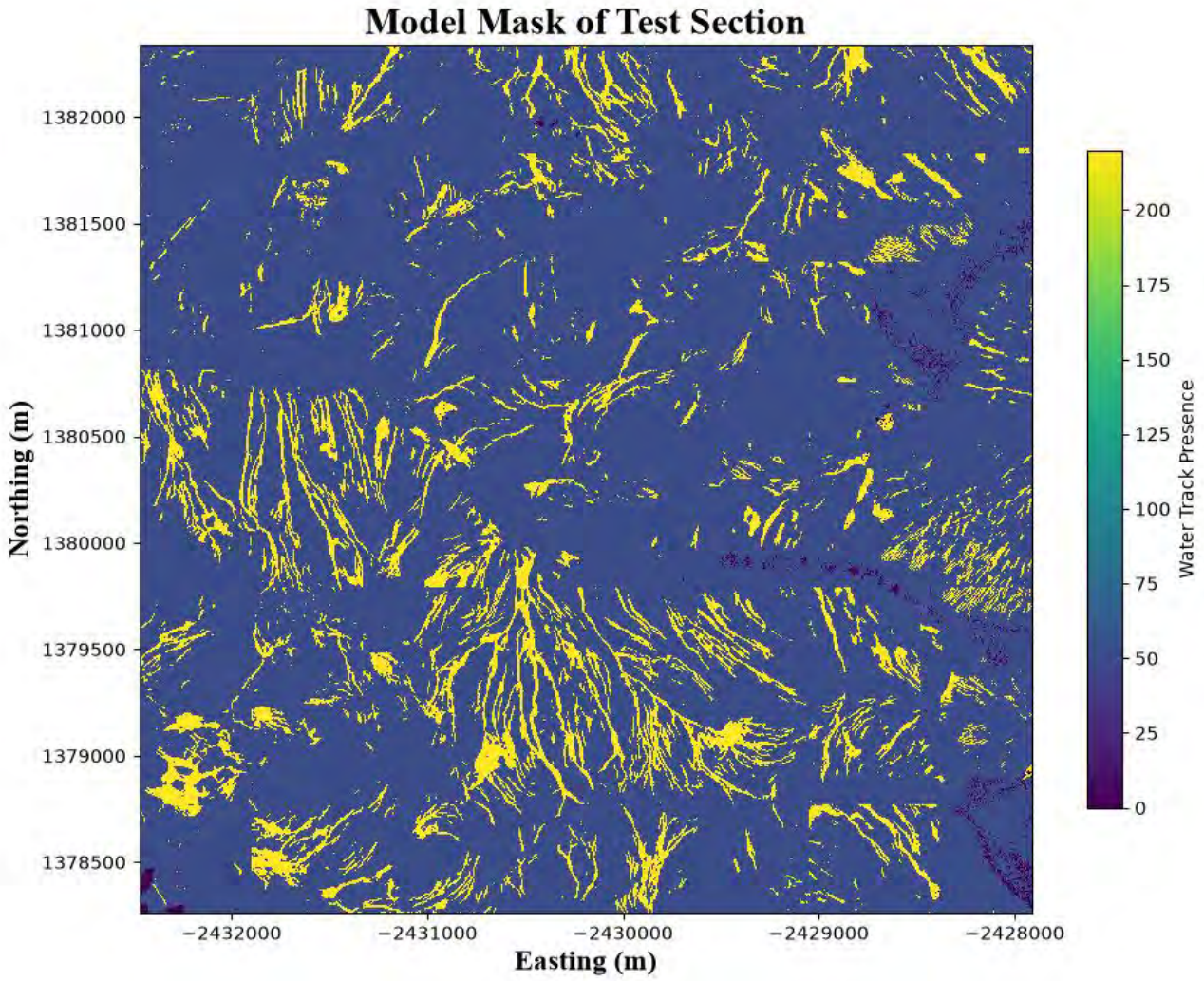
Binary recognition approach: So now Gabriel only has to classify **~5,000 tiles** to compare computer vision “eyeball” method to student “eyeball” method

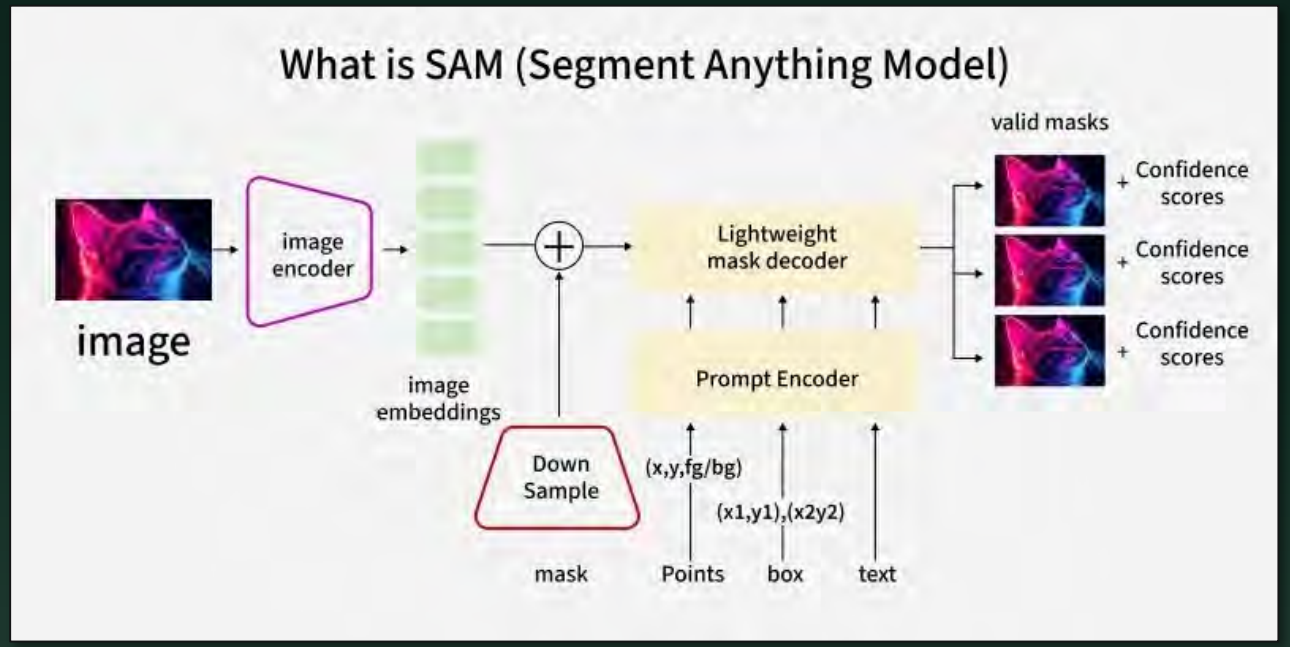




Water Track Segmentation Output on RGB Section from Vantor ID 1030010081A41700

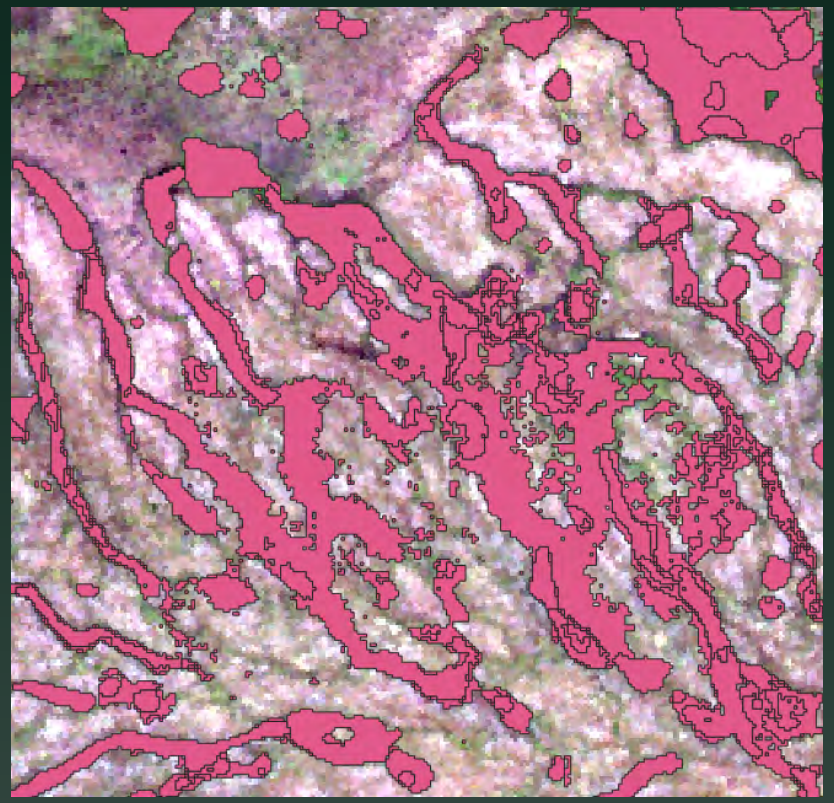




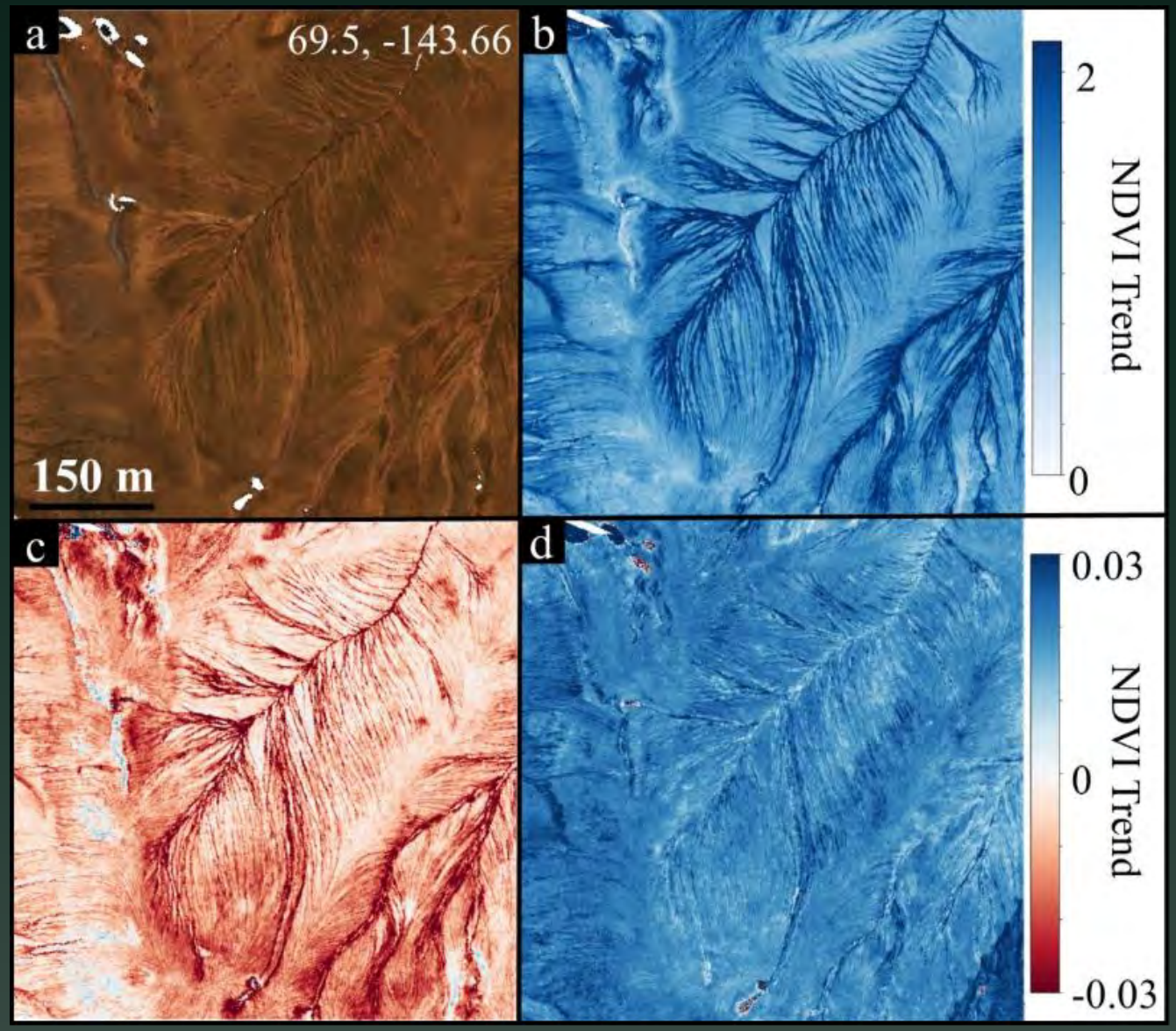


samgeo

Text prompt: “green lines”

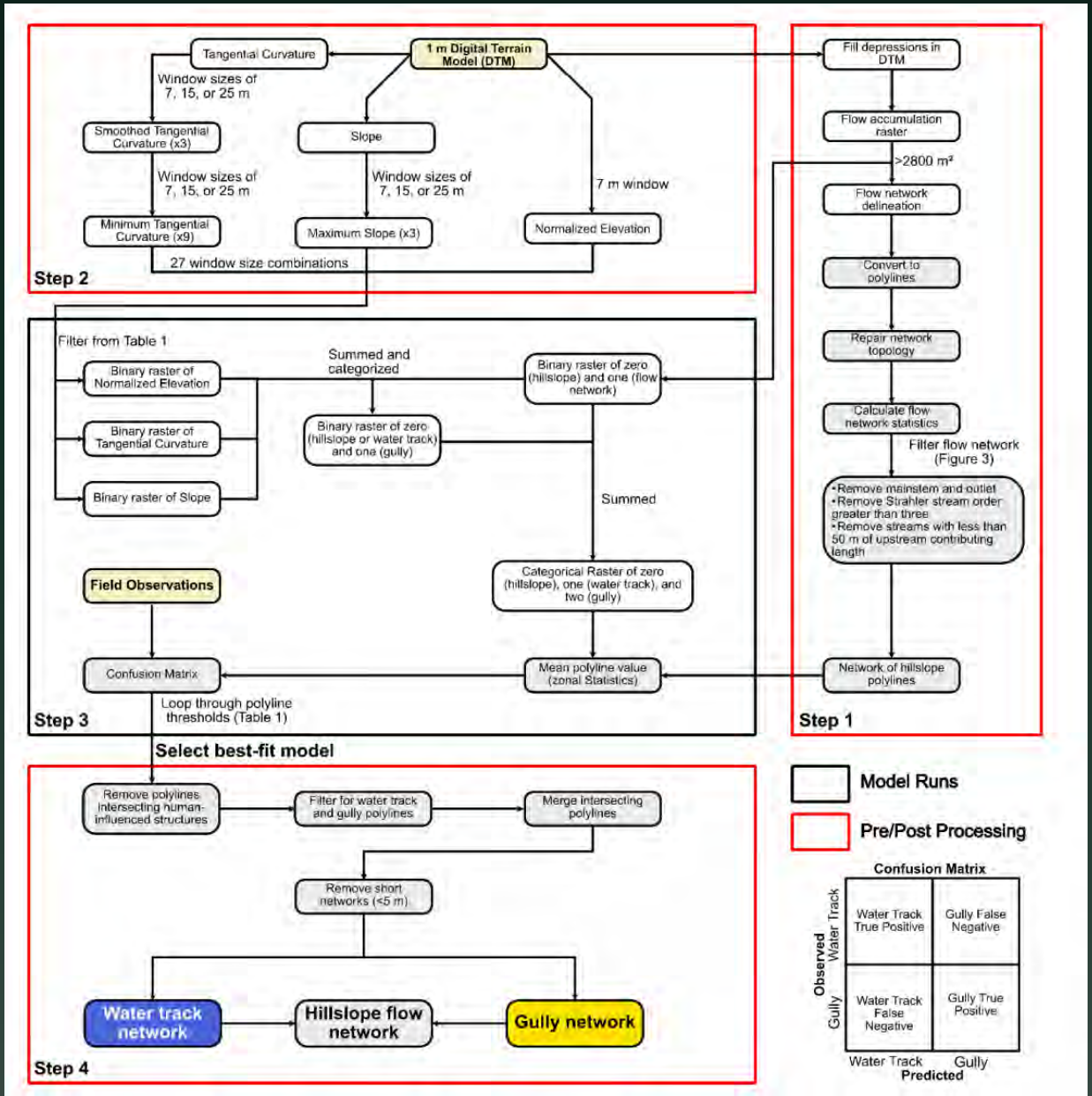


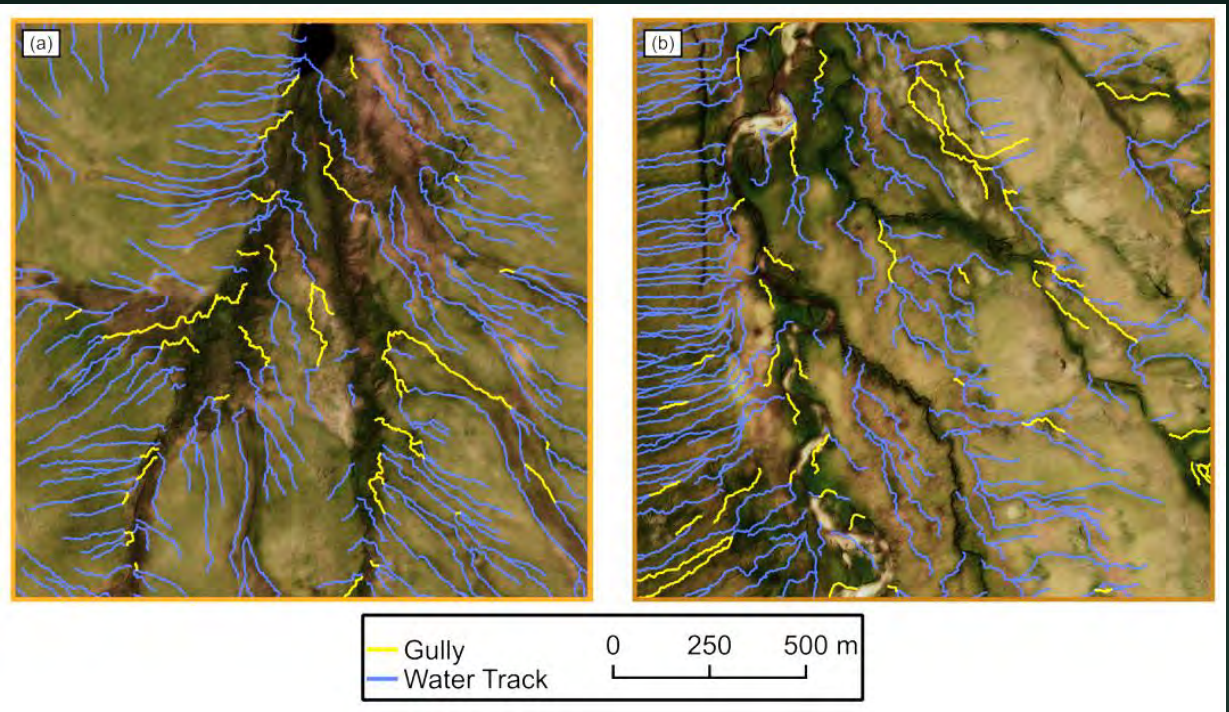
Aayla Kastning WM ‘26



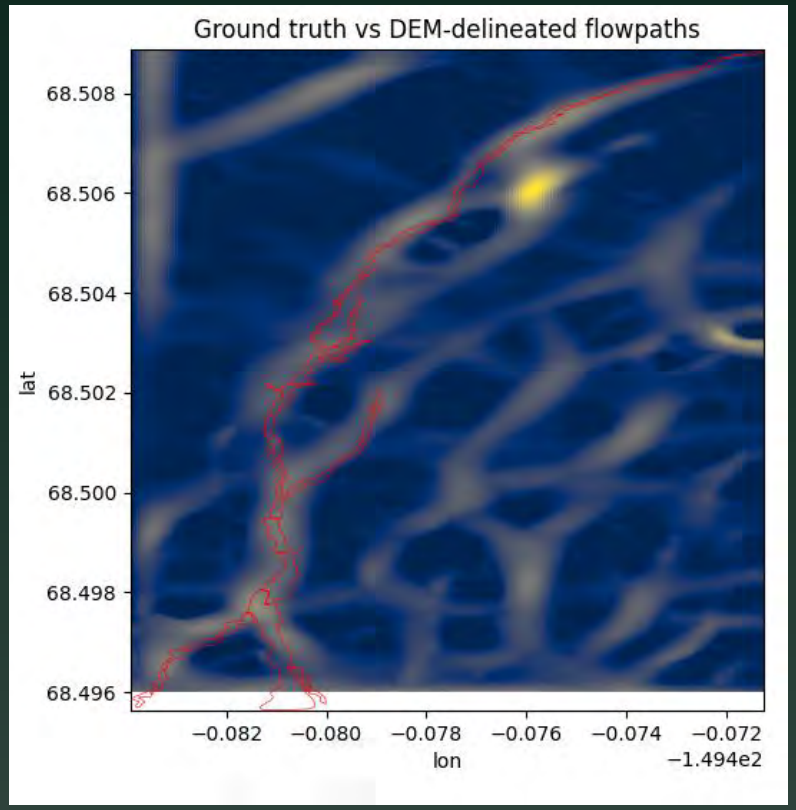


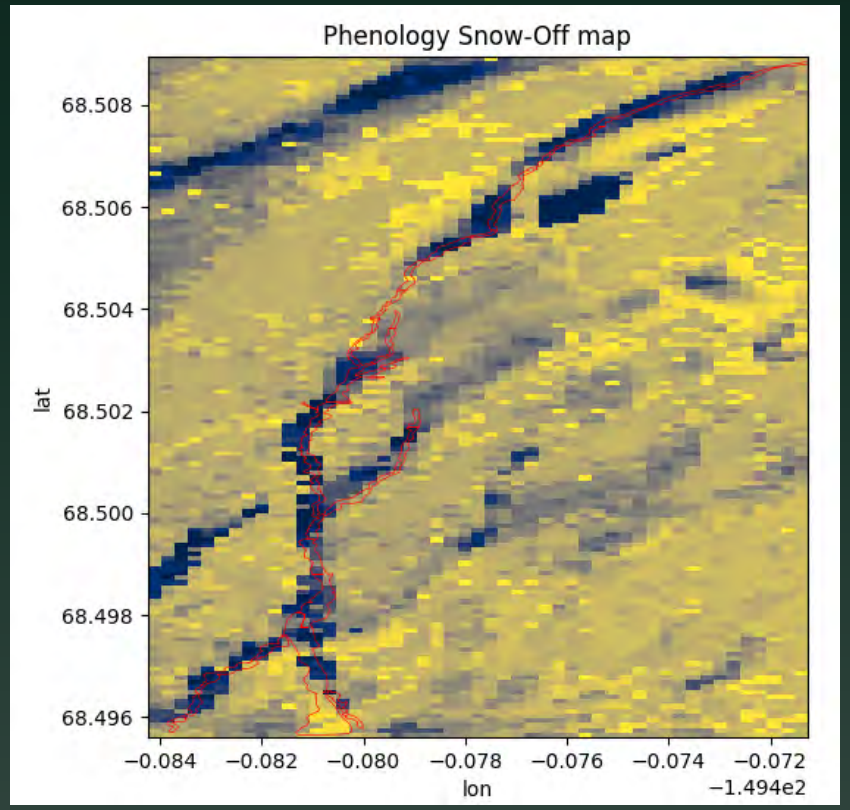


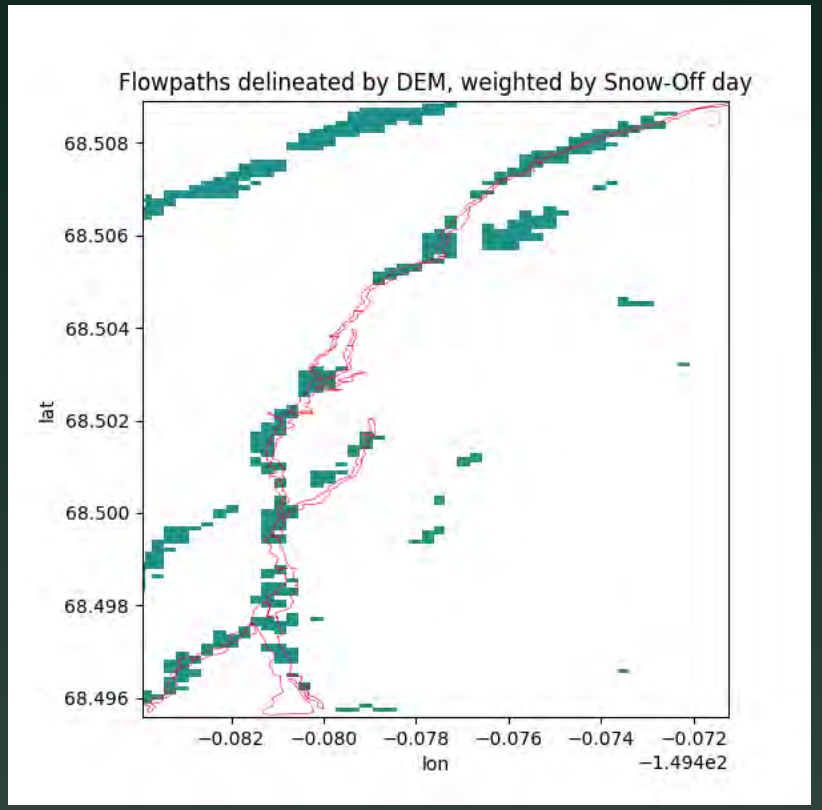


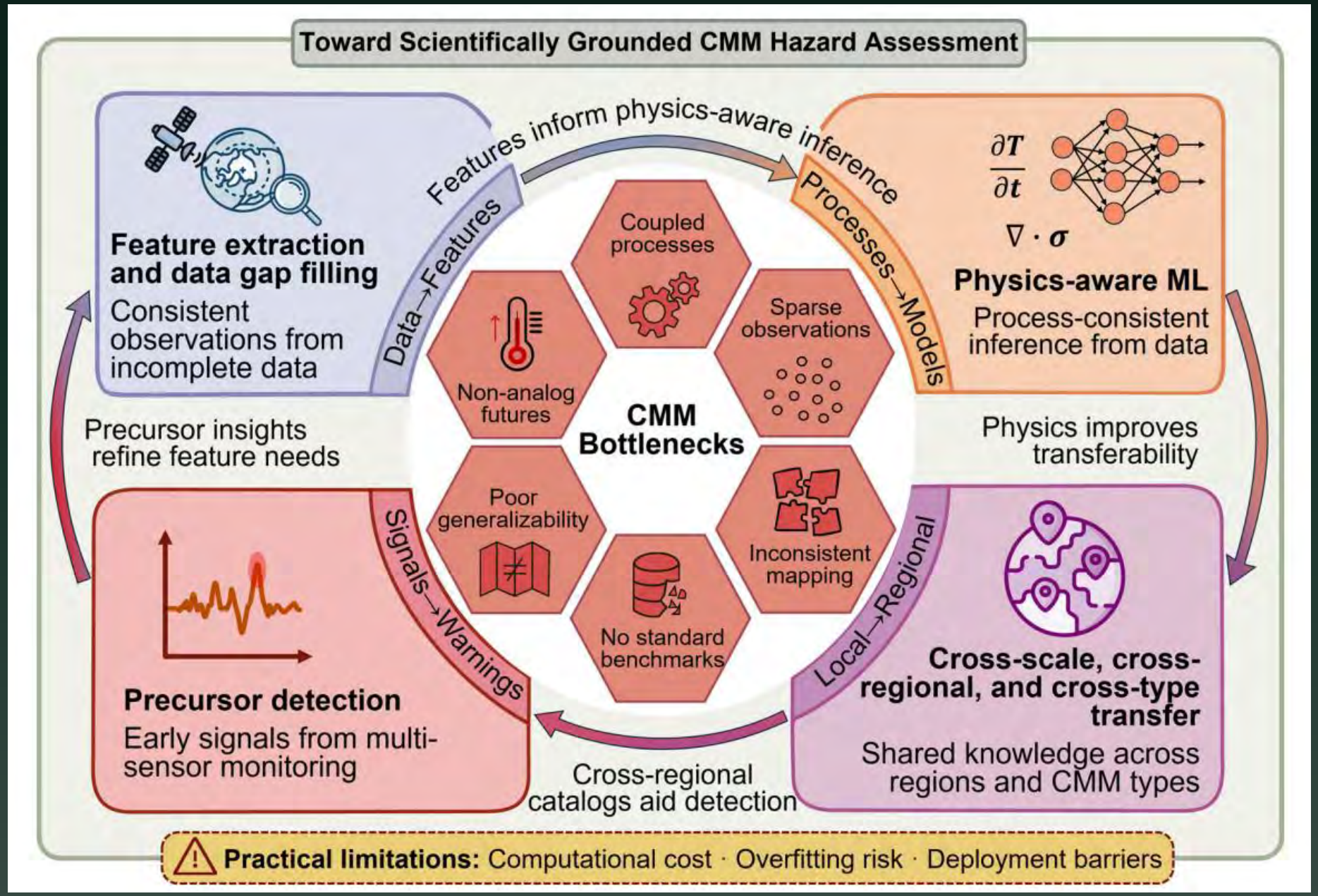


Bradon Yokeley (Idaho State)


Ground truth vs DEM-delineated flowpaths
68.508
68.506
68.504
68.502
68.500
68.498
68.496
lat
−0.082
−0.080
−0.078
−0.076
−0.074
−0.072
lon
−1.494e2

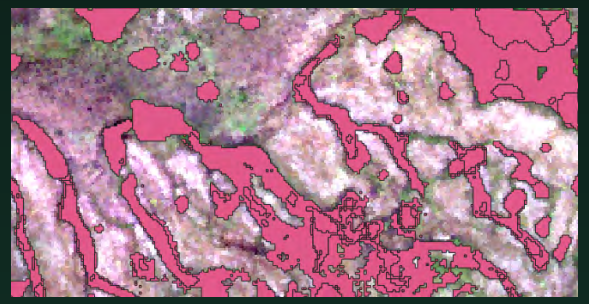

Phenology Snow-Off map
68.508
68.506
68.504
68.502
68.500
68.498
68.496
lat
−0.084
−0.082
−0.080
−0.078
−0.076
−0.074
−0.072
lon
−1.494e2


Flowpaths delineated by DEM, weighted by Snow-Off day
68.508
68.506
68.504
68.502
68.500
68.498
68.496
lat
−0.082
−0.080
−0.078
−0.076
−0.074
−0.072
lon
−1.494e2

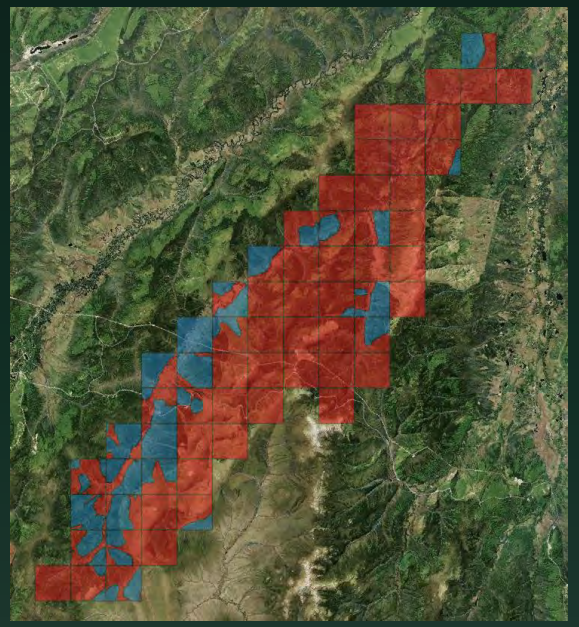


Pei, Te, Louise Farquharson, Margaret Darrow, Tong Qiu, Chaopeng Shen, Yingli Tian, Helena Bergstedt et al. "Machine learning for cryospheric mass movements: challenges and pathways." *npj Natural Hazards* 3, no. 1 (2026): 47.









Process (physics)-based geomorphology provides framework for predicting climate effects beyond historical observations

To what degree should maps and predictive models be **deterministic?**

As landscapes respond to **novel** climate drivers, how do we avoid imparting **bias** from training data?







joanmarie@wm.edu